

Market Access and Urban Growth in the Former Soviet Union^{*}

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Abstract

How does geopolitical fragmentation reshape the economic geography of cities? I exploit the 1991 Soviet collapse, which generated opposite market access shocks across regions. Using difference-in-differences on 1,396 cities spanning 1970–2021, I show that in post-Soviet Europe, cities near newly enforced borders experienced annualized population declines of 0.4–0.6 percentage points relative to interior cities, implying cumulative contractions of roughly 15%. In Central Asia, cities near previously sealed frontiers grew about 1 percentage point faster. Effects attenuate with distance, are amplified for manufacturing-intensive cities, and survive matching on pre-1991 characteristics. The findings point to market access as the primary channel.

JEL classification: F15, N94, O18, P25, R11, R12

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1 Introduction

Geopolitical fragmentation has redrawn national boundaries across the developing world in recent decades, creating new states from Eritrea to South Sudan while secessionist pressures simmer from Cameroon to Kashmir. Yet despite the frequency of these ruptures, causal evidence on how border changes affect urban growth in developing countries remains scarce. The empirical challenge is twofold. First, border realignments are usually shaped by local political and economic forces rather than occurring randomly. Second, such ruptures come bundled with other disruptions, making it difficult to isolate the effect of the border itself from the broader turmoil that accompanies fragmentation.

This paper exploits the 1991 collapse of the Soviet Union as a natural experiment to causally estimate how border realignments affect the economic geography of cities. The administrative boundaries that separated the Soviet republics were drawn in the 1920s, primarily based on ethno-linguistic criteria and centralized administrative convenience, rather than the economic potential or growth trajectories of individual cities (Kramer and Smetana, 2013). For seven decades, these lines imposed minimal friction on trade and migration (Eichengreen, 1993; Suesse, 2018). The dissolution was driven by aggregate, macro-level crises such as the collapse of global oil prices, union-wide fiscal failures, and political upheaval in Moscow, rather than the localized economic conditions of individual cities (Gaidar, 2010). The collapse caught even seasoned scholars off guard (Fukuyama, 1992; Kotkin, 2001). Because the location of these borders was historically predetermined and the timing of their enforcement was driven by macro-political forces, the resulting spatial shock to market access is plausibly exogenous to the pre-existing economic trajectories of individual cities.

At its core, this paper addresses a fundamental question: how does the sudden collapse of a unified state affect the spatial variation in economic activity? In the Soviet context, the political fragmentation of the state is precisely what transformed low-friction administrative lines into hard international borders. This abrupt political rupture economically fragmented the deeply integrated Gosplan supply chains, severing established trade networks and imposing severe market access penalties on cities that unexpectedly found themselves on the periphery of newly independent nations (Djankov and Freund, 2002). It offers a further advantage for distinguishing mechanism. Existing studies examine episodes in which market access moves in the same direction for all treated locations at a given moment, whether the shock is division or integration (Redding and Sturm, 2008; Ahlfeldt et al., 2015; Brühlhart et al., 2018). German division lowered market access for border cities and reunification later raised it, but the two shocks arrive in sequence, so the aggregate environment differs across them. The Soviet collapse instead hardened some frontiers and opened others simultaneously, so the sign of the border shock varies across cities while union-wide transition shocks do not.

In post-Soviet Europe, newly enforced internal borders severed the dense industrial supply chains that Gosplan had constructed across republican boundaries, while the simultaneous collapse of COMECON¹ redirected former Eastern Bloc partners toward Western Europe. In

¹The Council for Mutual Economic Assistance (COMECON) was an economic organization comprising the Soviet Union and its socialist allies, including Poland, Czechoslovakia, Hungary, Romania, and Bulgaria.

Central Asia, the dissolution opened previously sealed frontiers with Afghanistan, China, and Iran, borders that had been closed or heavily militarized throughout the Cold War (Pomfret, 2010, 2021). While the Central Asian border sample is smaller and geographically concentrated, suggestive evidence indicates these cities grew rather than shrank. That border cities decline in Europe and grow along the external frontiers in Central Asia following the exact same geopolitical rupture and robustness to multiple sensitivity checks provide compelling evidence that market access is the primary operative channel.²

Figure 1 maps the sample and the classifications used throughout. The map draws internal borders in blue and external borders in black, marks each city by the type of frontier it lies within 75 km of, and shades the two regional groupings, so a reader can see which shock a given city was exposed to.³

I estimate difference-in-differences models comparing cities within 75 km of borders to interior cities, using population data for 1221 cities in post-Soviet Europe and 175 in Central Asia spanning 1970–2021. The long pre-period permits direct testing of parallel trends, and the three-decade post-collapse horizon captures persistent effects. I distinguish two border types: *internal borders*, the former inter-republican boundaries that became international after 1991, and *external borders*, the frontiers that were already international throughout the Soviet period. I estimate effects separately for each type relative to interior cities within each region.

The main finding is that post-Soviet cities in Europe near internal and external borders see their populations decrease at annualized rates of about 0.4 and 0.6 percentage points (pp), respectively. These effects accumulate into relative contractions of roughly 11–17% over three decades. To contextualize this magnitude, the complete bisection of Germany by the Iron Curtain reduced the relative population of West German border cities by roughly one-third over a forty-year period (Redding and Sturm, 2008). That the post-Soviet fragmentation generated effects roughly half as large highlights the severe spatial costs of severing integrated supply chains, even in the absence of a completely militarized barrier. On the other hand, cities near Central Asia’s external borders grew faster than those in the interior by about 1 pp per year, though the smaller sample warrants caution. Internal border effects in Central Asia are not distinguishable from zero and are significantly smaller than in Europe, consistent with the thinner cross-republican production linkages that ran through Central Asian cities before dissolution (Pomfret, 2021).

Three pieces of evidence support market access as the operative mechanism. First, treatment effects are concentrated in the immediate vicinity of borders and generally attenuate with distance in both regions, a spatial pattern that ethnic migration or Soviet military capital flight would not produce but that market access models predict (Hanson, 2005). Second, I

²Section 3 defines market access as the composite of demand access and supplier access, the two dimensions of proximity to markets that jointly determine equilibrium city population. I do not construct a structural index (e.g., Donaldson and Hornbeck, 2016). Such an index is impractical here because Gosplan’s centralized material balance system violates the competitive market assumptions underlying standard gravity frameworks (Eaton and Kortum, 2002), and parameterizing the heterogeneous trade costs of newly hardened post-1991 borders is not feasible.

³I group the 15 republics into Europe (Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Russia, and Ukraine) and Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan) based on regional market access and economic geography (Council of the European Union, 2009).

exploit sectoral heterogeneity in exposure to cross-border linkages: manufacturing-intensive cities, more dependent on the vertically integrated supply chains that spanned republican boundaries, experienced amplified losses where borders hardened in post-Soviet Europe. This effect is difficult to reconcile with alternatives but follows directly from differential trade exposure. Third, the results survive nearest-neighbor matching on pre-1991 city size, industrial employment and composition, and defense sector presence, ruling out the possibility that border cities were on different trajectories before dissolution.

This paper contributes to the new economic geography literature, which since [Krugman \(1991\)](#) has emphasized market access as a central determinant of the spatial distribution of economic activity ([Helpman, 1998](#); [Fujita et al., 2001](#); [Redding and Venables, 2004](#)). Empirical work has tested this prediction through border or administrative frictions ([Behrens, 2024](#); [Egger et al., 2022](#); [Brühlhart et al., 2018](#); [Ahlfeldt et al., 2015](#); [Redding and Sturm, 2008](#)), transport infrastructure ([Jedwab and Storeygard, 2022](#); [Büchel and Kyburz, 2020](#); [Baum-Snow et al., 2017](#); [Storeygard, 2016](#); [Faber, 2014](#)), and trade policy ([Dix-Carneiro and Kovak, 2017](#); [Autor et al., 2013](#); [Hanson, 2001](#)). Closest to this paper, [Suesse \(2018\)](#) uses a gravity framework to show that Soviet disintegration severed inter-republican supply chains as nationalist mobilization advanced. My results complement that evidence. Suesse documents the trade collapse at the republic level; I trace its spatial incidence across individual cities, so the population responses estimated here rest on trade-based results already established at the aggregate.

This paper makes three primary contributions. First, it exploits a geopolitical rupture that simultaneously hardened some frontiers and opened others. This dual-shock setting generates identification from within-episode variation in the sign of the market access change, isolating the spatial friction of the border from the aggregate institutional collapse. Second, the massive scale of the empirical setting—tracking nearly 1,400 cities across 15 countries and dozens of border pairs over a five-decade panel—permits rigorous, well-powered tests of core theoretical mechanisms, including continuous distance gradients and sectoral heterogeneity. Third, it provides the first large-scale evidence on how the collapse of a centrally planned state reshapes urban systems, demonstrating that market access dynamics drive spatial divergence even in transition and developing contexts where these theories have rarely been tested.

More broadly, the findings offer city-level causal evidence for the economic costs of political fragmentation theorized by [Alesina and Spolaore \(1997, 2005\)](#), showing that these costs are spatially concentrated near newly created boundaries and vary with the pre-existing structure of cross-border economic linkages. The paper also speaks to research on the persistence of city sizes after major disruptions ([Miguel and Roland, 2011](#); [Brakman et al., 2004](#); [Davis and Weinstein, 2002](#)), providing novel evidence that border realignments can generate lasting spatial divergence when they durably rewire access to nearby markets.

These findings carry implications for contemporary geopolitics. Secessionist and autonomy movements are active on every continent, yet the economic costs of the borders they would create remain poorly understood, especially in developing countries. This paper shows that the consequences of fragmentation are neither uniform nor predictable from first principles: they depend on what economic linkages the new boundary severs and what opportunities it opens.

2 Institutional Background

The dissolution of the Soviet Union in 1991 fragmented one of the most deeply integrated economies in the modern world into 15 independent states. Mikhail Gorbachev’s *glasnost* and *perestroika* reforms were intended to preserve the system, not dismantle it (Kramer and Smetana, 2013), and few observers anticipated the total disintegration of the state (Smith, 1993). For spatial economics, this unexpected rupture constitutes a rare natural experiment.

Since its founding in 1922, the USSR organized a vast and diverse population into 15 Soviet Socialist Republics through national-territorial delimitation (*razmezhevanie*).⁴ Administrative borders between republics imposed little friction on internal trade or migration (Eichengreen, 1993; Denisenko and Chudinovskikh, 2017). Republican agencies retained limited planning autonomy and could on occasion influence allocation within their territory, but their remit extended to the republic as a whole rather than to particular locations within it. Gosplan, the State Planning Commission, coordinated production through hierarchical negotiations, deliberately constructing vertically integrated industries that spanned republican boundaries. Djankov and Freund (2002) estimate a gravity equation for trade among Russian regions and FSU republics and find that Russian regions traded no more with one another than with other republics before the collapse. A pronounced domestic bias emerged afterward as inter-republican trade contracted, with Russian regions trading roughly 60% more among themselves than with the republics.

To the extent that spatial frictions existed in the Soviet economy, they were organized around individual cities rather than administrative lines. For example, the *propiska* system, a residency permit regime introduced in 1932, regulated internal migration by restricting access to certain high-status cities. Critically, however, republican borders did not function as barriers to migration; their role was primarily administrative (Light, 2012; Gang and Stuart, 1999).⁵

2.1 Soviet Europe

Soviet republics in Europe were deeply integrated through centrally planned supply chains and infrastructure networks. The European USSR contained the most industrialized economies of the Union with factories often relying on single suppliers located in other republics for critical inputs. There was also far denser industrial and commercial integration in the European region of the USSR than in Central Asia before dissolution (Pomfret, 2021). Blanchard and Kremer (1997) demonstrate that when central planning’s enforcement mechanism disappeared before market institutions emerged, bargaining broke down across these supply chains. Output fell farthest for goods with the most complex production processes, precisely because they depended on cross-republican coordination. The post-Soviet transition transformed administrative lines into

⁴Most of the territory of the USSR had been part of the Russian Empire, including the Baltic provinces, which were Russian until 1918 and independent between the wars before annexation in 1940. The principal exceptions are territories acquired in and after 1940: western Ukraine (formerly Habsburg Galicia and interwar Poland), northern Bukovina, Transcarpathia, and the Kaliningrad exclave.

⁵Large urban centers, such as Moscow, Saint Petersburg, and Tashkent, were subject to especially strict *propiska* controls (Buckley, 1995). The results are robust to excluding such cities from the sample.

hard international borders. Transportation corridors were severed, production chains disrupted, and labor flows curtailed.

On the western frontier, Soviet external trade with Eastern Europe was organized through COMECON, which accounted for roughly half of Soviet foreign trade.⁶ Founded in 1949, COMECON bound the Soviet Union to Eastern European satellites through administered pricing and bilateral clearing arrangements that insulated members from global competition (Holzman, 1985).

COMECON effectively dissolved on January 1, 1991, when members shifted to hard currency transactions. The collapse reflected converging pressures: the 1989 revolutions removed the political rationale for Soviet-centered integration, and declining oil revenues eliminated Soviet capacity to sustain subsidies (Gaidar, 2010). Intra-bloc trade dropped from 13 times gravity-predicted levels in 1990 to just 4 times by 1993 (Maurel and Cheikbossian, 1998).

2.2 Central Asia

Central Asian republics occupied a distinct position in the Soviet economy. Functioning primarily as suppliers of raw materials (cotton, natural gas, minerals) rather than participants in industrial supply chains, they were integrated with Russia through extractive north-south corridors but lacked the dense inter-republican linkages characteristic of European USSR (Pomfret, 2021). All roads, railways, and pipelines led north to Russia. Central Asian economies therefore had fewer complex supply-chain relationships running through the boundaries that hardened in 1991.

Before 1991, Central Asian republics faced near-total isolation from neighboring non-Soviet states. The Sino-Soviet border was closed in 1962 and remained heavily militarized through the Cold War (Radchenko, 2009). The first Central Asian rail link not routed through Russia opened only in 1990 (Pomfret, 2021). The Iran-Turkmenistan border saw minimal contact, with all external trade channeled through Moscow. The Afghanistan border became a war zone following the 1979 Soviet invasion.

The dissolution opened all three external frontiers simultaneously. The Afghanistan-Tajikistan border, previously a war zone, and others saw the emergence of cross-border bazaar economies as shuttle traders (*chelnoki*) carried goods across newly accessible crossings (Spector, 2017). The Iran-Turkmenistan border, through which Moscow had previously channeled all contact, reopened to direct bilateral trade. The Khorgos crossing on the Kazakhstan-China border reopened in 1992, with approximately 700,000 Kazakhstani citizens crossing to China that year for “shop tourism” (Peyrouse, 2008). These new connections meant Central Asian border cities experienced market access gains.

⁶Trade with Türkiye and Iran remained minimal throughout the Cold War. The Armenian-Turkish border stayed closed after 1991 due to the Nagorno-Karabakh conflict. Türkiye and Azerbaijan share only a 17 km land border, offering little commercial potential in the early post-Soviet years, and Georgia’s early-1990s wars further disrupted the frontier.

Figure 2 documents the reorientation of post-Soviet trade.⁷ Panel A shows that European republics sharply reduced trade with former Eastern Bloc partners beginning in January 1991, when COMECON shifted to hard-currency transactions, while Central Asian republics increased their trade share with China and Afghanistan. Panel B reveals that all three regions expanded their non-FSU trade shares, though rising shares could reflect either genuine reorientation or simply the collapse of the FSU trade denominator.

Panel C addresses this by showing total interrepublic trade in constant 1990 rubles. The decline is steep but gradual, unfolding over several years as national currencies, customs barriers, and the breakdown of the ruble zone emerged sequentially across 1991–1994 (Gaidar, 2010). By 1994, interrepublic trade had fallen to roughly one-fifth of its 1990 level. Panel D illustrates divergent trade balances: Russia maintained surpluses throughout the period, while Central Asia shifted from substantial deficits to near balance by 1994, reflecting diminished capacity to import manufactured goods from former Soviet partners.

The asymmetry reflects fundamental differences in pre-1991 trade composition. Soviet planning assigned Central Asian republics the role of primary commodity suppliers while over 80% of industrial output concentrated in Russia and the western republics (IMF et al., 1991). European republics hosted vertically integrated supply chains with inter-republican trade flows often exceeding those between Canadian provinces (Michalopoulos and Tarr, 1996).

The nature of the market access gained in Central Asia also differed qualitatively. European border cities had been integrated with high-income partners on both sides: other Soviet republics internally and COMECON members externally. When these borders hardened, former COMECON partners reoriented toward the EU, compounding the loss. Central Asian border cities gained access to lower-income neighbors: Afghanistan, Iran under sanctions, and China’s least developed western provinces. As Kaminski and Mitra (2012) and Karrar (2019) document, the resulting cross-border trade was dominated by cheap consumer goods for resale in domestic bazaar economies, requiring little fixed investment and allowing gains to materialize rapidly through informal shuttle trade.

The Soviet collapse also triggered significant migration, particularly among ethnic minorities. Roughly a million ethnic Germans and Jews emigrated to Germany, while ethnic Russians returned to Russia from other republics, driven by economic instability, political uncertainty, and social pressures (Becker et al., 2012; Heleniak, 2004). A natural concern is whether ethnically motivated resettlement, rather than market access, drives the estimated border effects.

Several pieces of evidence suggest they do not. The dominant flows were ethnically motivated repatriations to ancestral homelands, not localized sorting away from borders (Mukhina, 2007; Heleniak, 2004). The largest emigrant populations concentrated in major interior metropolises, yet large cities show no significant border effects while small cities bear the full treatment weight. Treatment effects along internal borders also exhibit smooth spatial decay consistent with trade-cost models rather than discrete jumps at ethnic boundaries.

⁷The series ends in 1994, when Soviet-format interrepublic accounting was discontinued; the figure therefore establishes the abruptness and regional asymmetry of the initial shock rather than the long-run path of trade, which I discuss below.

3 Conceptual Framework

This section sets out the spatial equilibrium logic that organizes the analysis. I draw on the economic geography models of [Krugman \(1991\)](#), [Helpman \(1998\)](#), and [Fujita et al. \(2001\)](#), in the multi-region form used by [Redding and Sturm \(2008\)](#) and extended to intermediate inputs following [Redding and Venables \(2004\)](#). I use the framework to sign and interpret reduced-form estimates rather than to estimate structural parameters.

The mechanism requires that labor and goods be allocated by markets, and before 1991 they were not. Gosplan assigned production administratively, with little regard to the transport costs that shape market outcomes, and the *propiska* system regulated mobility by city status rather than by position relative to a frontier. A city's size therefore did not generally depend on its distance to any border. The framework predicts no differential response to border proximity before 1991, and the flat pre-trends in [Figures 3 and 4](#) are consistent with that prediction. [Redding and Sturm \(2008\)](#) report the corresponding check in their setting: re-estimating for centrally planned East Germany yields a division treatment close to zero. The treatment here is thus the onset of market allocation interacting with frontiers that simultaneously hardened and opened.

Workers are mobile and consume traded varieties and a local amenity in fixed supply. Firms produce differentiated varieties under monopolistic competition using labor and a composite intermediate, and both final and intermediate goods bear iceberg trade costs $\tau_{ic} > 1$. The price index of the intermediate bundle is the same trade-cost-discounted aggregate that governs the consumer cost of living, so a city's marginal cost is decreasing in its access to suppliers. Two distinct objects therefore summarize a city's position in this network. Supplier access is the trade-cost-discounted measure of varieties a city can obtain as inputs,

$$SA_c = \sum_i n_i (p_i \tau_{ic})^{1-\sigma}, \quad (1)$$

and demand access is the trade-cost-discounted measure of expenditure its firms can reach,

$$DA_c = \sum_i (w_i L_i) P_i^{\sigma-1} \tau_{ci}^{1-\sigma}, \quad (2)$$

where n_i is the mass of varieties produced in i and P_i its price index. Spatial equilibrium requires indirect utility to be equalized across populated cities, which yields equilibrium population as

$$L_c = \kappa H_c SA_c^\gamma DA_c^\eta, \quad \gamma, \eta > 0, \quad (3)$$

where H_c is the local amenity stock and γ and η depend on σ , the expenditure share on traded goods, and the intermediate share in production.

Supplier access and demand access are distinct economic objects but not distinct empirical ones. Both are sums over the same bilateral trade costs, so a frontier that hardens lowers both. Because the outcome I observe is population rather than prices or wages, [equation 3](#) recovers the composite and cannot separate them. I use *market access* for that composite throughout. In the

Soviet case the supplier margin is likely the larger, since output was allocated administratively while input deliveries were not readily substitutable (Blanchard and Kremer, 1997), but the design does not identify the decomposition.

The sign of the population response depends on the direction in which 1991 moved τ for a given city, weighted by the share of that city's input purchases and output sales crossing the frontier in question. Where enforcement raised trade costs on a frontier carrying a large share of a city's linkages, equation 3 predicts decline; where a previously prohibitive frontier opened, it predicts growth. Opposite-signed regional effects therefore follow from one comparative static rather than separate theories, and the regional difference is a share parameter rather than a difference in institutions or planning regime. I test this with interactions for Central Asia.

For external frontiers the treatment is not a change in τ but the onset of the regime in which τ matters. These boundaries were already close to closed, but under administrative allocation a truncated hinterland carried no penalty. Once allocation became spatial, cities at the edge of the union found much of the radius around them occupied by another economy. What equations 1 and 2 discount is economic mass rather than realized trade, and in post-Soviet Europe both components of the mass available to these cities moved against them. On the domestic side, the frontier truncated the radius within which a city could reach suppliers and buyers once distance began to bind.

On the foreign side, the two blocs had been each other's dominant partners: intra-bloc trade ran near half of COMECON members' total trade, and Eastern Europe drew the largest single share of its imports from the Soviet Union and the Soviet Union in turn from Eastern Europe (Holzman, 1985), with gravity estimates placing intra-bloc flows at thirteen times predicted levels in 1990 (Maurel and Cheikbossian, 1998). Those partners reoriented toward Western Europe after 1991, so the mass reachable across the frontier fell even as the frontier itself opened. The framework therefore predicts decline at external borders, and partial recovery where accessible foreign mass subsequently grew, as with EU accession in the Baltics, which I do not test directly. Redding and Sturm (2008) find reunification effects far smaller than division effects, consistent with slow adjustment of the population stock.

In Central Asia the comparative static runs through the other margin. A lower share of production there ran through inputs sourced across republican boundaries, so enforcing internal frontiers severed fewer supplier links and the framework predicts a smaller internal effect, though individual republics departed from the regional average. What the opened external frontiers supplied was buyers, in the bazaar economies documented by Kaminski and Mitra (2012), and buyers enter DA_c rather than SA_c . The European internal-border decline is therefore primarily a loss of supplier access and the Central Asian external-border gain primarily an addition of demand access, both components of the composite in equation 3. What determines exposure is not sector as such but the share of a city's purchases and sales crossing the frontier whose τ changed. Manufacturing employment tracks that share in Europe. In Central Asia it does not, since manufacturing there faced north toward Russia rather than toward the frontiers that opened, and low agricultural employment proxies instead for orientation toward cross-border commerce.

Beyond the sign, the framework predicts that effects should be localized, decaying with distance from the changed frontier, since cities farther away already faced high trade costs to locations across it. What decays is the hinterland a city forgoes, which depends on the geometry of the frontier and on city size, so the decline need not be strictly monotone in straight-line distance; [Redding and Sturm \(2008\)](#) find their own 0 to 25 km coefficient smaller than their 25 to 50 km coefficient. Effects should also be larger for small cities, since own-city expenditure enters DA_c at $\tau_{cc} = 1$ and is therefore a larger share of a large city’s total access, so a given loss of external mass is proportionally smaller.

Population adjusts through internal migration. As access falls the real wage falls, and workers leave until utility is re-equalized across cities. Migration is the mechanism by which the shock is realized rather than a rival account of it. The competing hypothesis is narrower, namely ethnically motivated resettlement, which would track the geography of minority settlement rather than distance to a frontier and would concentrate in the large cities those populations left.

4 Data and Empirical Methodology

The empirical analysis draws on the following data sources:

1. Population census. The main variable of interest is population growth. The dataset includes a panel of 1396 cities in the former Soviet Union, encompassing the period from 1970 to 2021. It covers the populations of cities that had over 5,000 residents in 1970. This selection criterion ensures that the sample includes all cities whose demographic composition was less likely to be affected dramatically by the dissolution.⁸ City population data with coordinates was collected from [Brinkhoff \(2021\)](#), while data from 1970 and later years, if missing, was gathered from Demoscope Weekly, an electronic database of the Russian Institute of Demography.⁹ City population data from before 1991 come from the Soviet-wide censuses conducted in 1970, 1979, and 1989. For the post-1991 sample, data were compiled for three distinct periods, with similar intervals for each country. A breakdown of the census data by country is presented in [Table A1](#).

I use the annualized growth rate of population following [Redding and Sturm \(2008\)](#), as it allows comparing growth rates consistently across different census periods, adjusting for length variations in those periods.¹⁰

2. Night lights. Sub-national GDP data is not available across the FSU. Instead, I use high-resolution nighttime lights data from the Defense Meteorological Satellite Program (DMSP), spanning 1992 to 2013, as a proxy for city-level economic activity. [Henderson et al. \(2012\)](#)

⁸The results are robust to using a threshold of 10,000.

⁹Available at [citypopulation.de](#) and [demoscope.ru](#), respectively.

¹⁰It can be computed in city c as

$$y_{ct} = \frac{\ln(P_{ct}) - \ln(P_{c,t-L_{ct}})}{L_{ct}}, \quad (4)$$

where P_{ct} is the population statistic at time t , and L_{ct} is the number of years between t and the prior census. It is the rate at which a population grows on average per year over a specific period, measured in log differences. Following [Redding and Sturm \(2008\)](#), I annualize to account for varying intercensal intervals. Log differences are symmetric, additive across periods, and nearly identical to the discrete compound annual growth rate at the typical growth rates in this sample (mean 0.68%).

validate this approach, showing that nighttime light growth closely tracks GDP growth in low- and middle-income countries with poor subnational income data.¹¹

3. Industry data from 1989. The paper uses total industrial employment, turnover, manufacturing output in the civilian industry, and the SIC code of each city’s primary sub-industry from 1989, compiled by [Kofanov and Mikhailova \(2015\)](#) from Soviet statistical yearbooks. Because city names in their dataset are recorded in Cyrillic and often reflect post-Soviet renaming, I match cities to the main sample using transliterated name strings and manually resolve remaining discrepancies using oblast (province) identifiers, historical city name registries, and coordinate matching. This procedure successfully matches 1,225 of the 1,396 cities in the full sample (1,076 of 1,221 in Europe and 149 of 175 in Central Asia). The unmatched cities are predominantly small places whose names changed after independence or whose Cyrillic transliterations do not correspond uniquely to a single entry in the population panel. Cities with no reported sectoral employment are also absent, which may skew the analysis toward cities with more complete industrial records.

4. Defense industry data. Data on Soviet defense enterprises comes from [Dexter and Rodionov \(2024\)](#), who compiled a comprehensive database of military factories and research establishments across the USSR. The raw database contains over 33,000 enterprise records spanning the full Soviet period. I restrict the sample to enterprises whose military work ended between 1989 and 1991, the window surrounding dissolution, and whose workforce size is documented (categories 1 through 3 on the database’s ordinal scale). Enterprises that ceased military production before 1970 or began after 1991 are excluded. Because city names in the defense database do not map directly to the population panel, I match locations using a combination of automated string matching and manual resolution, successfully linking 822 in the filtered database to cities in the main sample. For each matched city, I construct two variables: the count of defense enterprises that stopped work in the 1989–1991 window, and the sum of their ordinal workforce sizes.

Appendix [Table A2](#) provides descriptive statistics for the main estimation panels separately for Europe and Central Asia. [Table A3](#) provides summary statistics of the variables by region.

Empirical Strategy

The core identification strategy compares cities located near borders to cities in the interior. The identifying variation comes from the differential change in population growth experienced by border cities relative to interior cities across all post-Soviet countries, after versus before the dissolution. This cross-country pooling is appropriate because all cities within each region operated under the identical, highly centralized Soviet institutional framework prior to 1991. In the baseline specification, city fixed effects absorb all time-invariant differences across cities and period fixed effects absorb shocks common to all cities in a given round. In the more demanding robustness specification, sub-region-by-period fixed effects replace common period fixed effects, restricting the comparison to border versus interior cities *within the same sub-region and census period* (Baltics, Caucasus, Eastern Europe, and Central Asia) and absorbing sub-region-specific macroeconomic trajectories.

¹¹The data cleaning procedure is discussed in detail in Appendix A1.

What identification requires is weaker than the absence of friction. Whatever autonomy republican agencies exercised operated at the level of the republic and applied alike to its interior and its border cities, so it gives no reason for cities near administrative boundaries to follow different growth trajectories from interior cities in the same country.

The two border types differ in what the 1991 rupture changed. At internal frontiers it raised trade costs where they had been minimal. At external frontiers, as Section 3 sets out, trade costs were already high and what changed was the allocation regime that makes them bind. Both are tested against interior cities within the same region and period.

Formally, the DiD approach does not require that border locations were randomly assigned but assumes that, absent the sudden dissolution, trends in the outcome variable would have been parallel between border and non-border cities (Roth et al., 2023; Meyer, 1995).

While spatial spillovers from border to interior cities technically violate SUTVA (Rubin, 1980), capturing this relative divergence is the goal of the DiD framework, consistent with spatial equilibrium models (Fujita et al., 2001; Crozet, 2004; Redding and Sturm, 2008).

European border cities lost market access (both internally and externally), while Central Asian border cities gained it at external frontiers. This cross-regional contrast strengthens the case for market access as the operative channel, since alternative explanations would need to generate opposite-signed effects across regions to account for the full pattern of results. The causal estimates within each region, however, do not depend on across-region comparisons; they rely on border-versus-interior variation, which the sub-region-by-period specification further tightens to within-sub-region comparisons.

To validate the parallel trends assumption and examine dynamic effects over time, I employ an event-study framework. This approach allows for the assessment of pre-treatment differences, potential anticipation effects, and heterogeneous impacts across different post-dissolution periods. I examine how treatment effects evolve over time and detect any potential pre-trends with the help of the following specification:

$$y_{ct} = \alpha_c + d_t + \sum_{k=T_0}^{-2} \delta_{1,k} [IB_c \times D_k] + \sum_{k=0}^{T_1} \delta_{1,k} [IB_c \times D_k] + \sum_{k=T_0}^{-2} \delta_{2,k} [EB_c \times D_k] + \sum_{k=0}^{T_1} \delta_{2,k} [EB_c \times D_k] + \epsilon_{ct} \quad (5)$$

where y_{ct} is the annualized growth rate of population in city c in census period t ,¹² IB_c takes a value of one for cities within 75 km of an internal border and zero otherwise,¹³ EB_c takes a value of one for cities within 75 km of an external border and zero otherwise, and D_k is an indicator equal to one when the observation falls in event-time period k relative to the Soviet dissolution (with $k = 0$ denoting the first post-collapse census period) and zero otherwise.

Under this approach, the treatment effects are defined over an event-time window $k \in [T_0, T_1]$, which I fix to $[-2, +2]$. These effects are estimated relative to the excluded period immediately preceding the observed event (i.e. $k = -1$). For $k < -1$, the coefficients $\delta_{1,k}$ and $\delta_{2,k}$ capture potential pre-trend dynamics. Conversely, for $k \geq 0$, they measure the evolution of treatment

¹²It covers five intercensal periods – 1970-1979, 1979-1989, 1989-1999, 1999-2009, and 2009-2019 – with variations after the collapse depending on specific country census years.

¹³I also check the robustness of the results to different distance thresholds.

impacts following the enforcement of new borders and the dismantling of the Iron Curtain, respectively.

City fixed effects α_c absorb time-invariant observable and unobservable characteristics such as geographical location, historical infrastructure, and long-standing ethnic/cultural elements unique to each city, including any pre-division differences. Year fixed effects d_t control for time-varying shocks that are uniform across all cities, like economic shocks and secular trends in the growth rate of populations. To account for heteroscedasticity and autocorrelation, the standard errors are clustered at the city level.

Because post-Soviet countries conducted censuses at different dates, the time index denotes relative census periods rather than calendar years. The outcome is annualized to place growth rates on a common scale. I also report specifications with sub-region-by-period fixed effects that absorb group-specific trajectories, ensuring identification comes from border-versus-interior variation within each sub-region and period. Country-by-period fixed effects would be ideal but are infeasible: several small countries have too few interior cities to identify border effects within-country.

To strengthen causal interpretation, I conduct several robustness checks: varying the distance threshold for border proximity, excluding major cities subject to *propiska* restrictions, dropping countries one at a time, and including country and border-pair fixed effects. I also rule out alternative explanations using nearest-neighbor matching. Each matching exercise pairs border cities with the two closest interior cities in the cross-section based on a single pre-1991 covariate (1970 population, 1989 industrial employment, turnover, manufacturing output, SIC classification, or defense sector presence), estimated via probit propensity scores with matching performed with replacement. The matched sample retains all treated cities and their nearest neighbors, then re-estimates the baseline DiD on the restricted panel. Because matching is with replacement, a popular control city may serve as a neighbor for multiple treated cities but enters the estimation sample only once.

5 Results

DiD Estimates

Table 1 begins the analysis with the pooled sample of all 1,396 cities. Columns (1)–(2) impose a common border effect across regions. Column (1) estimates annualized declines of 0.21 pp for cities near internal borders and 0.563 pp for cities near external borders. Because smaller republics such as the Baltic and Caucasian states have mechanically higher shares of border cities, common period fixed effects could conflate national trajectories with border effects. Column (2) therefore replaces them with sub-region-by-period fixed effects. The external border effect is nearly unchanged at -0.540 pp, significant at the 1% level, while the internal border effect falls to -0.159 pp and loses significance. This attenuation reflects the homogeneity these columns impose: a common coefficient averages a negative European effect against a Central Asian effect that, as the next columns show, is indistinguishable from zero.

Columns (3)–(4) relax that restriction by interacting the treatments with a Central Asia indicator. The IB and EB coefficients now report effects for post-Soviet Europe, at -0.397 and -0.591 pp under period fixed effects and -0.241 and -0.663 pp under sub-region-by-period fixed effects. The triple interactions report the Central Asian differences, and a joint test rejects the restriction that both equal zero ($p = 0.006$ and $p = 0.002$). The Central Asia main effect is 0.079 pp ($p = 0.807$), so interior cities in the two regions experienced statistically indistinguishable changes in growth after 1991. The regional contrast is thus specific to border cities rather than a difference in aggregate regional trajectories.

The implied Central Asian effects reported at the foot of the table show where that difference lies. Internal border effects are 0.411 and 0.471 pp and are not distinguishable from zero ($p = 0.256$ and $p = 0.201$), consistent with the limited inter-republican linkages that existed before dissolution (Pomfret, 2021). External border cities grew 0.781 pp ($p = 0.074$) and 0.969 pp ($p = 0.031$) faster than interior cities, implying a relative cumulative population increase of roughly one-third over three decades. This sample is small and geographically concentrated: 11 of the 15 external border cities lie along the Afghanistan–Tajikistan frontier, with two near the Iran–Turkmenistan border and one near the Kazakhstan–China border. The consistency of positive point estimates across these distinct segments is reassuring, but the estimates warrant caution.

Columns (5)–(6) estimate the two regions separately. Because Central Asia constitutes one of the four sub-regions, the interaction specification in Column (4) already estimates region-specific effects: its implied Central Asian effects reproduce Column (6) exactly, and its European coefficients reproduce a Europe-only regression with sub-region-by-period fixed effects. The separate analysis is therefore a special case of the pooled specification rather than an alternative to it.

The event-study estimates that follow validate the parallel-trends assumption underlying these specifications and trace how the effects evolve across census periods.

Event-Study Estimates

Figure 3 shows non-parametric population trajectories for cities near internal (a and c) and external borders (b and d), respectively, compared to interior cities, with separate panels for Europe and Central Asia. Each subfigure indexes total city population to 1970 levels and plots the difference between border and interior indices as a basic DiD. Notably, all four panels exhibit parallel trends between border and interior cities prior to 1991, lending support to the identification assumption. In Europe, cities near internal borders experienced a sharp and sustained decline relative to interior cities after 1991. The pattern for external borders is even starker: European cities near these frontiers declined sharply as former COMECON partners reoriented trade toward Western Europe. Central Asia presents a contrasting picture. Descriptively, cities near internal borders show a relatively faster growth trajectory after 1991, while cities near external borders experienced even larger gains from newly opened access to Chinese and other markets previously closed under Soviet rule.

Figure 4 presents event-study estimates from equation 5, with $k = -1$ as the omitted reference period. Panel A shows results for Europe, Panel B for Central Asia. In Europe, the pre-collapse coefficient at $k = -2$ is close to zero for both border types, supporting the parallel trends assumption. In Central Asia, internal border pre-trends are similarly flat, while external border cities show a negative coefficient at $k = -2$ that is marginally significant, suggesting some caution in interpreting the Central Asian external border results.

In Europe, both border types show sharp and persistent declines after dissolution, with effects peaking at $k = 1$ and remaining strongly negative through $k = 2$. For external borders, this timing aligns with the progressive reorientation of former COMECON partners toward the EU. For internal borders, the delayed peak is consistent with Blanchard and Kremer (1997)’s account of cascading supply chain failures. The persistence of effects three decades after dissolution indicates lasting spatial divergence rather than temporary disruption.

In Central Asia, the pattern reverses. Internal border effects remain small and statistically insignificant across all post-dissolution periods, consistent with limited inter-republican trade to disrupt (Pomfret, 2010). External border cities show a gradual emergence of positive effects. The coefficients at $k = 0$ and $k = 1$ are positive but imprecisely estimated, consistent with the initially informal and uneven nature of cross-border commerce as shuttle traders began exploiting newly opened frontiers (Kaminski and Mitra, 2012). By $k = 2$, the effect becomes significant at the 5% level, suggesting that the gains from border opening materialized as trade corridors formalized over the longer run. The pre-collapse coefficient at $k = -2$ for external borders is negative and marginally significant, which warrants some caution; however, this single pre-period coefficient may reflect noise in the small Central Asian external border sample (75 clusters) rather than a systematic pre-trend.

Figure A1 confirms that the event-study patterns are robust to replacing period fixed effects with sub-region-by-period fixed effects in Europe and to excluding Turkmenistan in Central Asia. The external border effects in Europe are, if anything, larger and more precisely estimated under this more demanding specification.

Robustness

First, because the two treatments are defined by proximity to different frontiers, a city can lie within 75 km of both. 80 European cities do, which is 56% of all European external border cities and 23% of internal border cities, and the baseline specification switches both indicators on for them. Table A4 therefore partitions cities into four mutually exclusive categories: internal exposure only, external exposure only, both, and interior. In Europe, internal-only cities decline by 0.358 pp ($p = 0.001$) and external-only cities by 0.441 pp ($p = 0.130$), while cities exposed to both decline by 1.099 pp ($p < 0.01$). A test that the dual effect equals the sum of the two single effects is not rejected ($p = 0.456$), so the two frontiers compound rather than interact. The external-only cell contains 62 cities and cannot reject zero on its own ($p = 0.088$), but the incremental effect of external exposure among cities already near an internal border is estimated more precisely, at -0.740 pp ($p = 0.007$), strengthening to -0.875 pp ($p = 0.001$) under sub-region-by-period fixed effects. I therefore read the external-only null as a limitation of the 62-city

sample rather than as evidence against an external border effect. The internal-only coefficient, by contrast, attenuates to -0.172 pp and is imprecisely estimated under sub-region-by-period fixed effects, indicating that within-sub-region identification of the internal border effect rests substantially on doubly exposed cities.

Columns (3)–(4) report the same decomposition for Central Asia, where it supports little inference. Only 5 cities lie near both frontiers, so the dual-exposure coefficient rests on a handful of places and none of the pairwise comparisons approaches significance. The external-only estimate is the one that holds up. It is 1.356 pp ($p < 0.05$) in Column (3) and 1.192 pp ($p < 0.10$) in Column (4), which excludes Turkmenistan, and both exceed the 0.969 pp pooled estimate in [Table 1](#). Isolating external from internal exposure therefore strengthens the Central Asian gain. The internal-only estimate moves the other way, from 0.552 pp and insignificant in Column (3) to 0.778 pp ($p < 0.05$) in Column (4).

[Table A5](#) splits the sample at the median 1970 city population, and the European effects are carried almost entirely by small cities (column 3). Below the median, internal border cities decline by 0.447 pp and external border cities by 0.587 pp, both significant at the 5% level. Above the median the estimates fall to negative 0.130 and 0.200 pp and neither is distinguishable from zero. The Central Asian external border gain follows the same pattern, with small cities growing 1.402 pp faster.

This is the size gradient predicted in Section 3, and [Redding and Sturm \(2008\)](#) report a comparable pattern for German division. The gradient also speaks to whether the estimates track post-1991 population movements. The ethnic populations that departed after dissolution came disproportionately from large urban centers, yet it is precisely there that border and interior cities are indistinguishable, and the divergence appears where those outflows were thinnest.

A potential inference concern is that ϵ_{ct} may be spatially correlated across nearby cities, particularly those sharing the same border segment. [Table A7](#) reports Conley spatial standard errors at cutoffs from 25 to 200 km ([Conley, 1999](#); [Colella et al., 2023](#)); all significance levels are preserved across both regions.

[Table A6](#) reports additional sensitivity checks. The results are robust to excluding large cities subject to *propiska* restrictions, which also addresses ethnic composition since these cities hosted disproportionately diverse populations and experienced the highest outmigration rates after 1991 ([Heleniak, 2004](#)) (Column 1); to dropping the turbulent 1989–1999 intercensal period, after which the European external border effect strengthens to 0.648 pp, consistent with progressive COMECON trade reorientation (Column 2). Column (3) removes cities within 75 km of both border types, so that the baseline specification is estimated on singly exposed cities alone; because these cities are dropped rather than assigned their own indicator, its coefficients reproduce the internal-only and external-only estimates in [Table A4](#) exactly, and the column is retained as a check on the headline specification in its original form.

Column (4) excludes conflict-affected border pairs (e.g., Armenia–Azerbaijan and Georgia–Russia). Removing conflict zones *strengthens* the European internal border effect to 0.690 pp ($p < 0.01$), confirming the baseline is not driven by populations fleeing violence. The external border effect remains marginally significant (-0.412 pp, $p = 0.071$).

Column (5) applies a region-specific restriction addressing how the regions are classified. In Europe it drops the 123 Russian cities in the Siberian and Far Eastern federal districts, confining the Russian sample to districts west of the Urals; the internal border effect strengthens to 0.489 pp and the external effect holds at 0.532 pp, both significant at the 1% level, so the European estimates do not depend on grouping Asian Russia with post-Soviet Europe. In Central Asia the column drops the ten cities on the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. The internal estimate rises from 0.471 to 0.569 pp and remains insignificant, a small movement but in the direction the mechanism implies, since what remains are borders that had less to sever.

Column (6) restricts the control group to cities within 750 km of any border. If the results reflected generic peripherality, narrowing the control group should attenuate both estimates. Instead the two border types respond asymmetrically in Europe: the internal effect strengthens to 0.529 pp while the external effect attenuates to 0.392 pp, though it remains significant at the 5% level. This is consistent with external border cities sharing structural features with the remote interior cities being dropped, while internal border cities lost linkages with nearby neighbors whose absence becomes more visible in the restricted sample. In Central Asia neither effect moves much, consistent with a compact sample offering less scope for peripherality to confound the estimates.

Figure A2 presents leave-one-out analyses. In Europe, both border effects remain negative and stable across all ten country exclusions; no single country drives the findings. In Central Asia, the external border coefficient remains positive across all five exclusions, but both effects are sensitive to dropping Kazakhstan, which contributes the largest share of interior control cities. With Kazakhstan removed, the remaining sample contains too few interior cities for precise estimation. The Central Asian internal border result should therefore be interpreted with caution, while the external border finding remains robust to the distance restrictions in Table A6, Columns 5–6.

Appendix Table A8 replaces the binary 75 km treatment with flexible 25 km distance bins, following Redding and Sturm (2008), to disentangle border-specific market access effects from generic peripherality. If the baseline results simply captured remoteness from national centers rather than proximity to newly enforced borders, effects should appear uniformly across all distance bins rather than concentrating near borders.

In Europe, internal border effects decay steadily with distance: the coefficient is largest at 0–25 km (a decline of 0.486 pp, significant at the 1% level), attenuates at 25–50 km (0.363 pp, significant at the 5% level), and fades to insignificance beyond 50 km. External border effects are strong and significant throughout the 0–100 km range, with the largest effect at 0–25 km (a decline of 1.055 pp). Under the more demanding sub-region-by-period specification (Column 2), the external border gradient strengthens, while internal border effects attenuate, consistent with the pattern in Table 1.

In Central Asia, the external border effect is concentrated in the closest bin: cities within 25 km of external borders grew 1.438 pp faster, significant at the 5% level, with coefficients positive but imprecise at greater distances. This sharp spatial concentration is consistent with

the localized nature of informal shuttle trade documented by Kaminski and Mitra (2012). The pattern of effects (broadly) decaying with distance from borders, rather than appearing uniformly across peripheral locations, supports the interpretation that the results reflect border-specific market access shocks rather than a general peripherality gradient.

Appendix B decomposes the aggregate effects into border-pair-specific estimates (Figures B1 and B2). Individual estimates are imprecise for pairs with few cities, but the broad pattern is informative: in Europe, the majority of internal border pairs show negative effects and the largest external border declines concentrate along former COMECON frontiers where trade reoriented westward. In Central Asia, effects are uniformly positive across all frontier segments. That the signs align with the aggregate results across geographically distinct borders in both regions provides additional support for the market access interpretation.

Figure C1 examines whether population results manifest in nighttime lights, a proxy for economic activity available from 1992. The cross-regional divergence in light intensity mirrors the opposing signs documented in the population regressions: European border cities show an initial relative decline while Central Asian external border cities display positive deviations through the early 2000s. A more detailed discussion is provided in Appendix C.

Sectoral Heterogeneity

The market access mechanism predicts that the magnitude of border shocks should vary systematically with a city’s pre-existing exposure to cross-border trade. Cities whose economies depended heavily on inputs, customers, or supply chain relationships spanning soon-to-be-political boundaries should experience larger disruptions when those boundaries harden. Conversely, when previously closed borders open, the gains should accrue to cities whose economic structure can exploit the newly accessible markets. I test this prediction using city-level data on 1989 industrial employment by sector, compiled by Kofanov and Mikhailova (2015) from Soviet statistical yearbooks.¹⁴ Because the distribution of manufacturing employment in Europe spans a wide range, I decompose the European border effect by manufacturing employment quartiles. In Central Asia, where the relevant sectoral contrast involves agriculture, I split cities at the within-region median. Appendix Table A10 reports median splits for all remaining sector-by-border-type combinations in both regions.

Table 2, Columns (1)–(2) present the European results. The mean manufacturing share is 0.834 with a median of 0.925, reflecting the heavy industrialization of the Soviet economy in Europe. A continuous linear interaction with such a compressed distribution is difficult to interpret, so I decompose the border effect by manufacturing quartiles, allowing each segment its own treatment effect without imposing a functional form. The results reveal a sharp threshold. Cities in the lowest manufacturing quartile near internal borders show no significant decline after border hardening (+0.305, $p = 0.216$), while all three upper quartiles experience declines of 0.5 to 0.8 pp that are individually significant and jointly strongly so ($p = 0.007$). Notably, the implied effects do not increase monotonically from Q2 to Q4. It reinforces the threshold interpretation:

¹⁴Summary statistics for the matched sample, including the distribution of cities across heterogeneity bins by border type, are reported in Appendix Table A9.

what matters is whether a city had substantial manufacturing at all, not the precise share. Cities in Q2 through Q4 all had manufacturing shares above 0.75 and were similarly embedded in cross-border production networks, so further increases in manufacturing concentration offered no additional exposure to the border shock.

The external border quartile gradient is directionally consistent, with Q4 cities declining nearly four times more than Q1 (-0.923 vs. -0.255 pp), but individual interactions are imprecise (joint test $p = 0.691$). This reflects both the smaller treated sample and the fact that the COMECON collapse imposed a broad market access loss on all external border cities regardless of sectoral composition, compressing the cross-quartile variation available for identification.¹⁵

Columns (3)–(4) turn to Central Asia. The relevant sectoral contrast here involves agriculture rather than manufacturing, for reasons of both economic logic and statistical power. The gains from border opening accrued through informal shuttle trade rather than industrial exchange (Kaminski and Mitra, 2012), so the key distinction is between commercially oriented cities and agricultural ones. Central Asian manufacturing was oriented northward toward Russia, not toward the frontiers that opened. The Central Asian EB sample contains only 11 treated cities; the manufacturing median split places just 2 in the high group against 9 in the low group, yielding no detectable difference ($p = 0.997$; Appendix Table A9). The agriculture split produces more balanced cells (8 high, 3 low among treated EB cities) and separates cities along the dimension that theory predicts should matter for this shock. The results confirm the prediction: low-agriculture EB cities grew 3.27 pp faster ($p = 0.006$), while cities with above-median agriculture showed no significant gains (0.647, $p = 0.324$), and the difference is statistically significant ($p = 0.047$). Internal border effects in Central Asia are uniformly insignificant across all sectors, consistent with the limited inter-republican trade that existed before dissolution.

Appendix Table A9 reports median splits for all remaining sector-by-border-type combinations. In Europe, the sharpest contrast involves mining. Cities with above-median mining employment show no significant decline at internal borders ($+0.059$, $p = 0.762$), while cities without mining decline by 0.578 pp ($p < 0.001$); the difference is significant at $p = 0.011$ and replicates at external borders (-0.170 vs. -0.915 pp, $p = 0.090$). This is economically intuitive: extraction operations are anchored to fixed natural deposits and do not depend on the cross-border supply chain linkages that border hardening severed. A city mining coal near a newly hardened border retains its productive base; a city assembling machinery from inputs sourced across republican boundaries does not.

In Central Asia, mining tells a different story. At external borders, cities with above-median mining grew 1.70 pp faster ($p = 0.047$), compared to 0.92 pp for non-mining cities ($p = 0.165$), though the difference is not statistically significant ($p = 0.453$). The point estimates are suggestive of mineral-exporting cities benefiting from newly accessible markets, but with only 1 treated city in the high-mining EB cell, the result cannot support strong inference. The Central Asian external border sample is small throughout (61 clusters, 11 treated), and the appendix results should be interpreted as indicative of patterns rather than as precisely estimated effects.

¹⁵These results are robust to including sub-region-by-period fixed effects, available upon request.

Taken together, the sectoral heterogeneity in both regions points in the direction that market access theory predicts: the cities whose economic structure was most exposed to the specific linkages created or destroyed by fragmentation experienced the most extreme effects.

Alternative Accounts

If Soviet central planning systematically concentrated specific industries or military assets in border regions or interior regions, the post-dissolution divergence could reflect compositional rather than border shocks. I address this through nearest-neighbor matching on pre-1991 city characteristics, testing whether border effects survive after restricting the sample to treated and control cities that were observably similar before dissolution.

Table 3 reports results for Europe from six matching exercises, each pairing border cities with interior controls based on a single pre-1991 covariate: 1970 population (Column 1), 1989 industrial employment (Column 2), turnover (Column 3), manufacturing output (Column 4), primary industrial classification (Column 5), and defense sector presence (Column 6).¹⁶ Across all six variables, internal border cities experience significant declines ranging from 0.36 to 0.58 pp, bracketing the baseline estimate of 0.40 pp. External border effects are similarly robust, with estimates ranging from 0.53 to 0.90 pp, all statistically significant. The consistency across matched samples that differ substantially in size and composition provides strong evidence that the European border effects are not artifacts of pre-existing differences in city characteristics.

Column (6) addresses military divestment specifically. The Soviet defense sector employed at least one in five industrial workers, and military budgets collapsed by over 90 percent between 1988 and 1998 ([Dexter and Rodionov, 2024](#)). I match on the count and workforce of defense enterprises that ceased operations between 1989 and 1991. The matched sample is smaller (267 cities for internal borders, 100 for external) but the border effects persist at similar magnitudes: -0.46 pp for internal borders and -0.83 pp for external borders, both statistically significant. Military capital flight therefore does not account for the relative decline of European border cities.¹⁷ The matching results for Central Asia ([Appendix Table A12](#)) should be interpreted with more caution: external border effects are positive in four of six columns, but the matched samples are very small (between 5 and 25 cities), precluding systematic inference.

Finally, another concern runs in the opposite direction from the one matching addresses. Soviet policy concentrated strategic R&D in a designated set of middle-sized cities, most of them specialized in nuclear physics, ballistics, aerospace, or chemistry, and state support for them was withdrawn abruptly after 1991: Russian gross R&D expenditure fell from roughly 2% of GDP in 1990 to 0.74% in 1992, and R&D employment halved ([Schweiger et al., 2022](#)).

¹⁶Matching is performed on the city cross-section using 1:2 nearest-neighbor propensity score matching with replacement for Columns (1)–(4) and (6). Each treated (border) city is paired with its two closest interior cities based on probit-estimated propensity scores. Because matching is with replacement, a control city may serve as a neighbor for multiple treated cities but enters the estimation sample once. Column (5) uses exact matching on SIC codes, retaining all border and interior cities that shared the same dominant industrial classification in 1989; the larger sample in this column reflects the retention of all cities within shared SIC strata rather than a fixed number of neighbors per treated city. Descriptive statistics confirming covariate balance in the matched samples are reported in [Appendix Table A11](#).

¹⁷A multivariate propensity score specification matching on all six covariates simultaneously yields highly similar estimates for post-Soviet European cities. The results are available upon request.

The natural worry is that such cities inherited a collapse of place-based support that my design would misattribute to border hardening. But Schweiger et al. (2022) show that the opposite happened. Comparing Science Cities to localities that were observationally similar at the time of their founding, they find that these cities outperform their matched counterparts continuously in population growth and in night lights from 1992 through 2010 with no visible catch-up by the control group. The authors attribute this to persistence interacting with agglomeration forces, estimating agglomeration elasticities around 0.3. For my design the implication is specific: because Soviet siting deliberately favored locations remote from the union’s frontiers, with about a third of these cities clustered around Moscow and a second group placed deep in the Urals and Siberia beyond the reach of enemy bombers, any such cities in my sample fall overwhelmingly in the interior control group. Cities that outgrew their peers for three decades would inflate the interior benchmark and make the estimated border penalties larger than they should be. Excluding them is therefore a test of whether the European results are manufactured by a handful of unusually resilient control cities.

Table A13 implements this. Schweiger et al. (2022) identify 95 Science Cities, 91 with recoverable locations. Matching by name recovers only a handful, since their unit is the Russian municipality rather than the city, many places were renamed after 1991 (e.g., Arzamas-16 became Sarov, Chelyabinsk-70 became Snezhinsk), and statistical sources were never published for closed cities on national security grounds.¹⁸ I therefore match on coordinates and exclude sample cities within a given radius of the nearest Science City, which flags the place itself where the two records coincide and its immediate neighborhood where they do not.¹⁹ Of the 91 locatable Science Cities, 62 have a city from my census sample within 10 km and 77 within 20 km. The exclusion removes 51 census sample cities at a 3 km radius and 102 at 20 km; the gap between the two counts widens with the radius because a single Science City can be the nearest neighbor of several sample towns, so the widest columns are best read as neighborhood exclusions.

Because the published list covers modern Russia alone, Columns (5)–(6) additionally exclude 20 closed or restricted-access cities elsewhere in the union, hand-coded from secondary sources. In Europe the internal border effect moves from -0.395 pp at 3 km to -0.365 pp at 20 km and the external effect from -0.609 to -0.592 pp, against baseline estimates of -0.402 and -0.595 pp; adding the closed-city exclusion leaves them at -0.361 and -0.593 pp on 1,112 of the original 1,221 cities. The slight attenuation runs in the direction the mechanism implies, since removing cities that outgrew comparable places lowers the interior benchmark. Only one matched Science City lies within 75 km of an internal border, so these places enter almost exclusively as controls. In Central Asia, where 13 cities are dropped, the external border gain attenuates from 0.969 to 0.833 pp and remains significant at the 10% level while the internal estimate is unchanged at 0.479 pp.

¹⁸Roughly a third were also built from scratch in sparsely populated areas and so fall below the 1970 population threshold defining my sample.

¹⁹Name matching fails asymmetrically rather than randomly, since the cities it misses are disproportionately the closed ones, for which Soviet and Russian agencies withheld statistics on national security grounds (Schweiger et al., 2022). Radius matching is also the right unit of comparison, because Schweiger et al. (2022) assign Science City status at the level of the municipality, a district-scale unit that may contain a Science City alongside other settlements, so a city-to-city join would leave those settlements in the control group.

6 Conclusion

This paper examines the effects of geopolitical fragmentation on urban growth, exploiting the Soviet collapse as a natural experiment. The setting offers unusual analytical leverage: identical institutional rupture producing opposite effects across regions with different pre-existing economic geographies, nearly 1400 cities across dozens of border pairs, and a five-decade horizon.

The central finding is that identical geopolitical fragmentation produced opposite effects depending on pre-existing economic geography. In post-Soviet Europe, cities near newly enforced internal and external borders experienced persistent population declines relative to interior cities. In post-Soviet Asia, cities near previously sealed external borders grew faster. The magnitude of these effects highlights the profound importance of market access for urban growth.

These findings carry implications beyond the post-Soviet context. Border changes are not mechanically economically harmful or beneficial for nearby places: their effects depend on what economic linkages the new boundary severs and what opportunities it opens. Where a border cuts through integrated production networks, nearby cities experience persistent relative decline; where it unlocks access to previously closed markets, nearby cities can experience sustained relative growth. Because the effects concentrate among smaller, more trade-dependent cities, geopolitical fragmentation widens spatial inequality within successor states, disproportionately penalizing cities least equipped to absorb the loss of cross-border market access. As movements for regional autonomy continue globally, this asymmetry deserves attention from policymakers weighing the spatial economic costs of fragmentation.

Figures & Tables



Figure 1: City Types and Regional Classification in the Former Soviet Union

Notes: The map shows all 1,396 cities in the sample. Cities are classified into four mutually exclusive categories by border exposure: within 75 km of a former inter-republican (internal) border only, within 75 km of an external border only, within 75 km of both, and interior cities beyond 75 km of any border. Blue lines mark internal borders, the boundaries between post-Soviet states that were administrative before 1991; black lines mark external borders, the frontiers that were already international throughout the Soviet period. Fills distinguish the two regional groupings used in the analysis: post-Soviet Europe comprises Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Russia, and Ukraine; Central Asia comprises Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan, following [Council of the European Union \(2009\)](#). Russia is classified with post-Soviet Europe despite spanning both continents: the grouping reflects the density of cross-republican production linkages rather than physical geography, and Russian cities in Asia are treated on the same basis as all others, facing internal frontiers with Kazakhstan and external frontiers with China and Mongolia. A small number of far-eastern Russian cities in the estimation sample lie outside the plotted extent. The 75 km cut-off follows [Redding and Sturm \(2008\)](#).

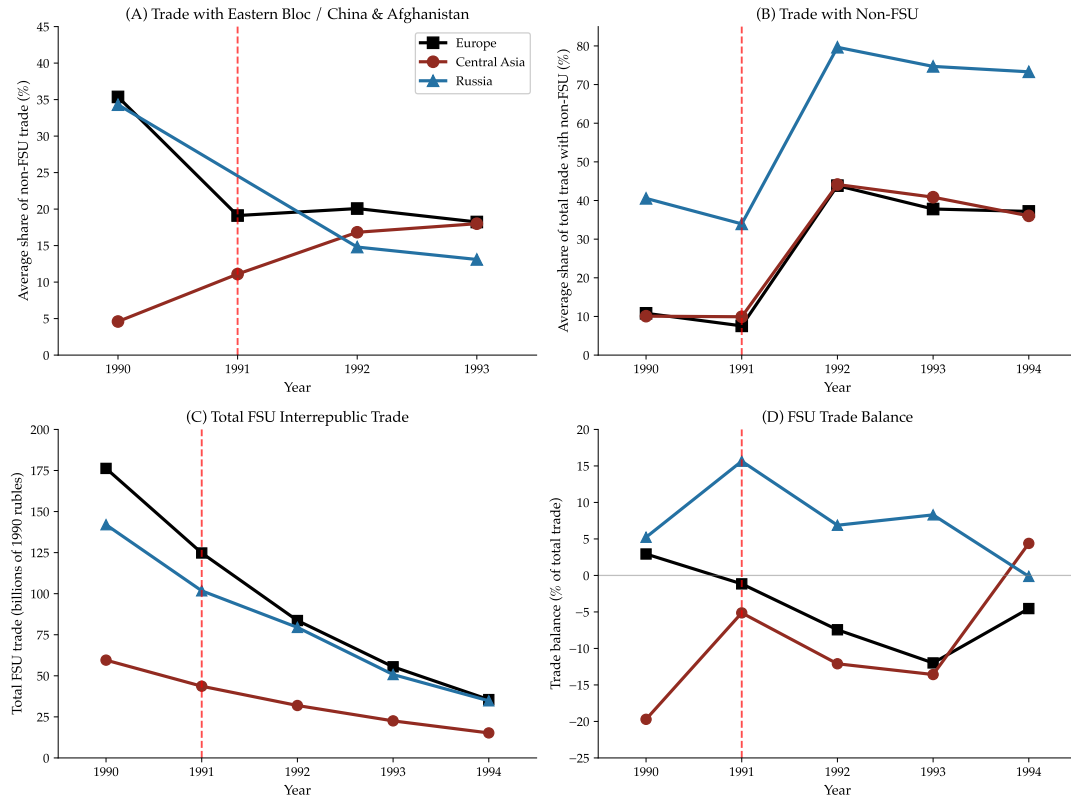


Figure 2: Post-Soviet Trade Realignment: Europe vs. Central Asia

Notes: This figure presents four panels illustrating trade reorientation patterns among Former Soviet Union (FSU) republics from 1990–1994. Countries are grouped into three regions: “Europe” (Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Moldova, Ukraine; $n = 9$), “Central Asia” (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Uzbekistan; $n = 5$), and Russia (shown separately). The vertical dashed line in each panel marks December 1991, when the USSR formally dissolved. **Panel A** shows the average share of non-FSU trade conducted with Bulgaria, Czechoslovakia, Hungary, Poland, and Romania for European republics and Russia, and China plus Afghanistan for Central Asian republics. Turkmenistan is dropped due to lack of data. Regional values represent unweighted country averages. **Panel B** shows the average share of each region’s total trade (exports plus imports) conducted with non-FSU countries (i.e., rest of world). This measure captures the overall reorientation away from intra-FSU trade toward global markets. Data are from IMF Direction of Trade Statistics and national sources. Regional values represent unweighted country averages. **Panel C** shows total interrepublic trade within the FSU, measured in billions of constant 1990 rubles to enable comparison across years despite hyperinflation that ensued. For each region, total trade equals the sum of exports and imports with all other FSU republics. **Panel D** shows the trade balance as a percentage of total FSU trade. Positive values indicate a trade surplus with other FSU republics; negative values indicate a deficit. Raw data source is [Belkindas and Ivanova \(1995\)](#). Coverage ends in 1994 because the Soviet-format interrepublic trade accounts on which Panels C and D rely were discontinued once the successor states began reporting as independent customs territories; no consistent constant-ruble series is available thereafter. The figure documents the sharpness and regional asymmetry of the initial reorientation rather than the trajectory of trade over the full study period.

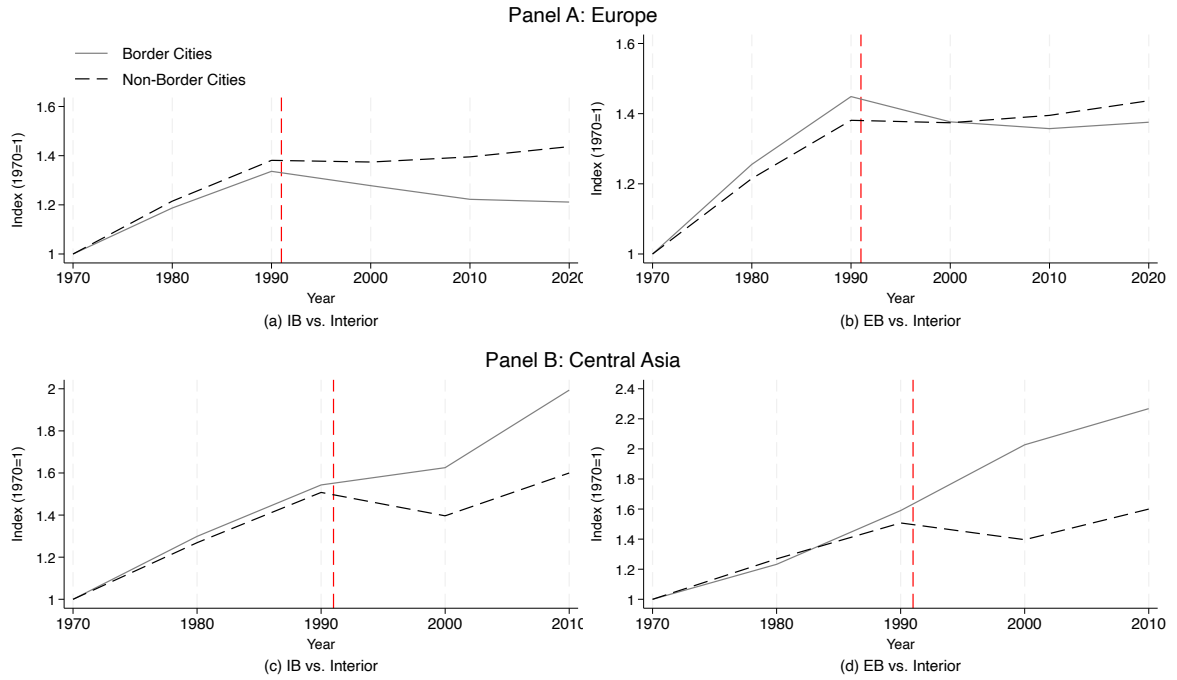


Figure 3: Indices of Border and Non-Border City Population

Notes: Panel A includes all post-Soviet countries geographically situated in Greater Europe: Armenia, Azerbaijan, Belarus, Estonia, Georgia, Latvia, Lithuania, Russia, and Ukraine. Moldova is excluded due to an unbalanced panel. The Central Asia panel includes all five post-Soviet Central Asian countries, but the balanced index shown here includes only two post-collapse periods, as Turkmenistan and Uzbekistan conducted only two censuses after the Soviet dissolution.

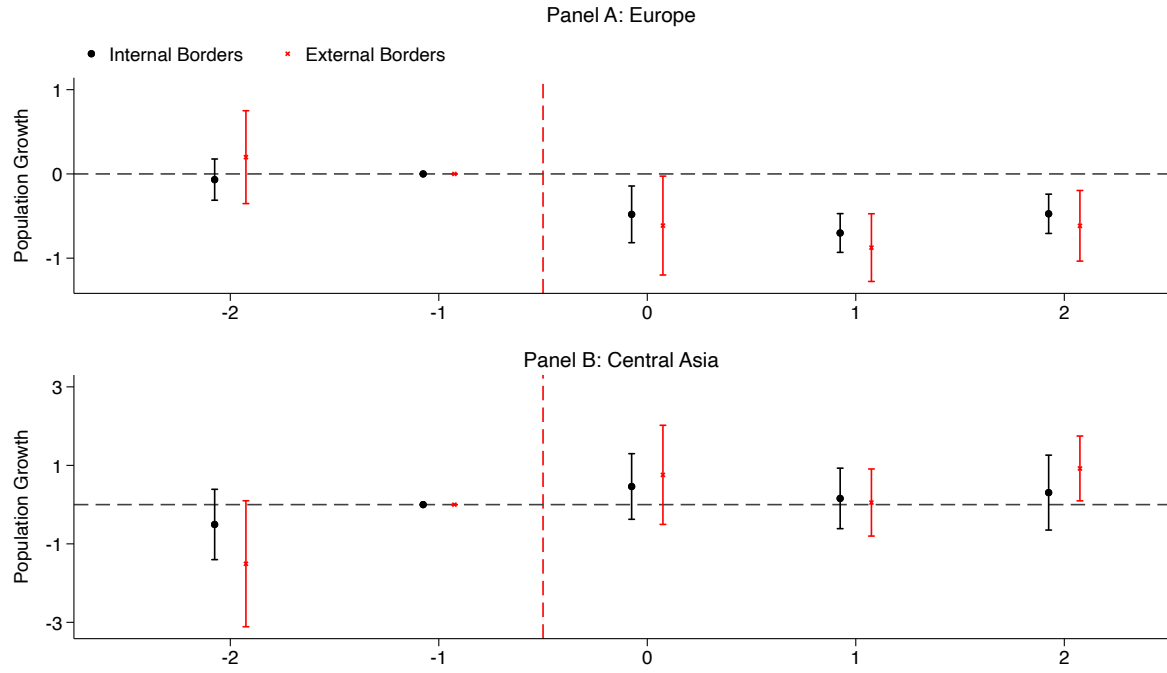


Figure 4: Event-Study Estimates: Baseline

Notes: Coefficient plots from event-study regressions of annualized population growth on border proximity interacted with census-period indicators (equation 5). Black circles denote internal borders; red crosses denote external borders. Panel A shows Europe; Panel B shows Central Asia. The omitted period is $k = -1$ (1979–1989). All specifications include city and period fixed effects. Control group: interior cities beyond 75 km from any border. Standard errors clustered at the city level; bars show 95% confidence intervals.

Table 1: Urban Growth and Market Access: Main Results

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
IB $\times$ Division	−0.210** (0.105)	−0.159 (0.112)	−0.397*** (0.111)	−0.241** (0.114)	−0.402*** (0.111)	0.471 (0.370)
EB $\times$ Division	−0.563*** (0.190)	−0.540*** (0.184)	−0.591*** (0.200)	−0.663*** (0.196)	−0.595*** (0.200)	0.969** (0.451)
CA $\times$ Division			0.079 (0.323)			
IB $\times$ Division $\times$ CA			0.808** (0.378)	0.712* (0.386)		
EB $\times$ Division $\times$ CA			1.373*** (0.480)	1.633*** (0.490)		
Implied IB effect, CA			0.411	0.471		
p -value			0.256	0.201		
Implied EB effect, CA			0.781	0.969		
p -value			0.074	0.031		
p : CA interactions jointly zero			0.006	0.002		
Outcome mean (%)	0.676	0.676	0.676	0.676	0.583	1.369
Sample	All	All	All	All	Europe	CA
Cities	1,396	1,396	1,396	1,396	1,221	175
Observations	6,809	6,809	6,809	6,809	6,004	805
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓		✓		✓	✓
Sub-Region-by-Period FE		✓		✓		

Notes: The dependent variable is annualized city population growth (%). “IB” and “EB” denote cities within 75 km of internal and external borders, respectively; “CA” denotes Central Asia (Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, and Uzbekistan). “Division” equals one for post-1991 periods. Columns (1)–(4) pool all 1,396 cities. Column (1) includes city and period fixed effects. Column (2) replaces period fixed effects with sub-region-by-period fixed effects, where sub-regions are the Baltics, the Caucasus, Slavic Europe plus Moldova, and Central Asia. Columns (3)–(4) add interactions between the Central Asia indicator and each treatment, so the IB and EB coefficients report effects for post-Soviet Europe and the triple interactions report Central Asian differences relative to Europe; the Central Asia main effect is collinear with the sub-region-by-period fixed effects and is therefore omitted in Column (4). Implied Central Asian effects are the corresponding coefficient sums, with p -values from Wald tests of the linear combination, and the final row reports a joint test that both Central Asian interactions equal zero. Columns (5)–(6) estimate the baseline specification on each region separately. Because Central Asia constitutes one sub-region, Column (4) is numerically equivalent to estimating the two regions separately with sub-region-by-period fixed effects: its IB and EB coefficients and its implied Central Asian effects reproduce those separate estimates exactly. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Urban Growth and Sectoral Composition

	Dependent Variable: Population Growth			
	Europe: Manufacturing Quartiles		Central Asia: Agriculture Median Split	
	(1) Internal	(2) External	(3) Internal	(4) External
Border $\times$ Division (Q1, ref.)	0.305 (0.247)	−0.255 (0.520)		
Border $\times$ Division $\times$ Q2	−1.102*** (0.335)	−0.313 (0.627)		
Border $\times$ Division $\times$ Q3	−0.912*** (0.334)	−0.659 (0.619)		
Border $\times$ Div. $\times$ Q4 (highest)	−0.813** (0.368)	−0.668 (0.738)		
Q2 $\times$ Division	0.460** (0.184)	0.460** (0.184)		
Q3 $\times$ Division	0.491*** (0.183)	0.491*** (0.183)		
Q4 $\times$ Division	0.474** (0.228)	0.474** (0.228)		
Border $\times$ Div. $\times$ High Agric.			0.554 (0.584)	0.647 (0.651)
Border $\times$ Div. $\times$ Low Agric.			0.340 (0.642)	3.268*** (1.135)
High Agric. $\times$ Division			0.273 (0.782)	0.259 (0.789)
<i>p</i> : joint Q2=Q3=Q4=0	0.007	0.691		
<i>p</i> : Q4 vs Q1	0.028	0.366		
<i>p</i> : High = Low			0.805	0.047
Outcome mean (%)	0.527	0.578	1.362	1.328
Cities	1,021	848	141	61
Observations	5,033	4,206	652	284
City FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). The matched industry sample contains 1,076 European cities. Columns (1)–(2) exclude cities near external borders only (*ext_only*=1), so that the internal border effect is identified against interior cities not exposed to external border shocks. Columns (3)–(4) similarly exclude cities near internal borders only. This accounts for the difference between the 1,076 matched cities and the column-specific sample sizes. Columns (1)–(2) decompose the European border effect by 1989 manufacturing employment quartiles, where Q1 (lowest manufacturing share) is the reference category. The quartile cutpoints are computed over all European cities with sectoral data. Column (1) estimates the internal border effect; Column (2) estimates the external border effect. Columns (3)–(4) split Central Asian border cities at the within-region median of 1989 agricultural employment share; “High” denotes cities above the median and “Low” denotes cities at or below the median. Because the median agricultural share is close to zero, the split effectively compares cities with any substantial agricultural employment to those without. “Border” refers to cities within 75 km of an internal border in Columns (1) and (3) and an external border in Columns (2) and (4). “Division” equals one for post-1991 periods. Columns (1) and (2) exclude cities near external and internal borders only, respectively. Standard errors clustered at the city level are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Robustness to Matching on Pre-1991 Characteristics (Europe)

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Internal Borders						
IB × Division	−0.576*** (0.128)	−0.361** (0.143)	−0.571*** (0.129)	−0.443*** (0.136)	−0.381*** (0.125)	−0.463** (0.199)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	0.568	0.542	0.457	0.471	0.512	0.537
Cities	737	655	651	657	945	267
Observations	3,601	3,207	3,187	3,217	4,654	1,309
Adj. R ²	0.427	0.432	0.433	0.436	0.454	0.474
Panel B: External Borders						
EB × Division	−0.528** (0.247)	−0.690*** (0.258)	−0.901*** (0.248)	−0.623** (0.252)	−0.626*** (0.237)	−0.826** (0.409)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	0.747	0.637	0.574	0.564	0.566	0.745
Cities	310	254	278	275	752	100
Observations	1,512	1,243	1,360	1,346	3,726	490
Adj. R ²	0.389	0.365	0.379	0.397	0.439	0.309
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. “Cities” in the matching tables reports the number of unique cities in the matched sample; “Observations” equals cities multiplied by census periods. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section: each treated (border) city is paired with its two closest interior cities based on probit-estimated propensity scores for the indicated covariate. Because matching is with replacement, a control city may serve as a neighbor for multiple treated cities but enters the estimation sample once. Column (5) uses exact matching on each city’s primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989; the larger sample reflects the retention of all cities within shared SIC strata rather than a fixed number of neighbors per treated city. “Military” matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). “Division” equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Appendix A - Figures & Tables

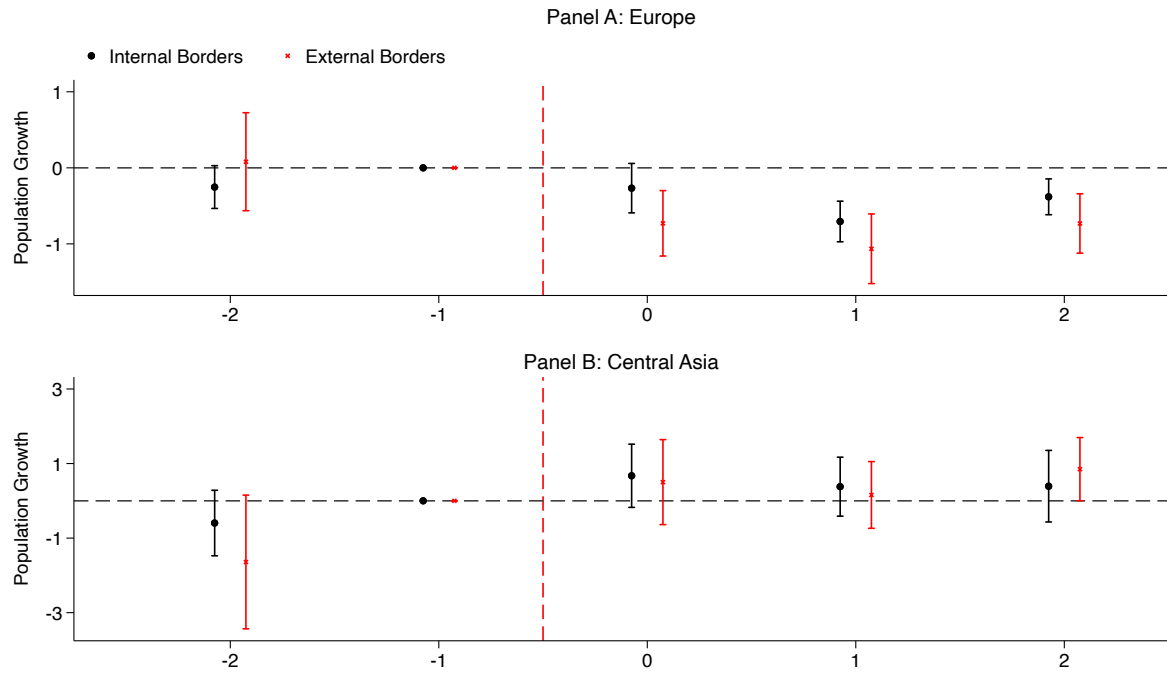
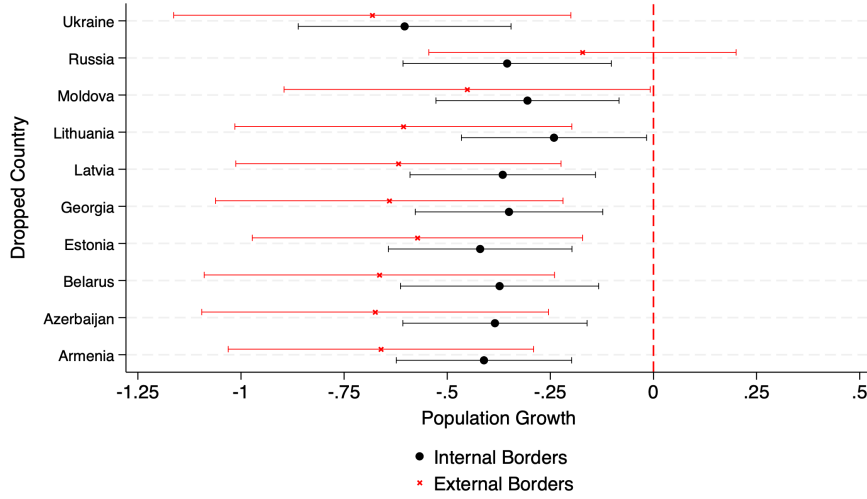


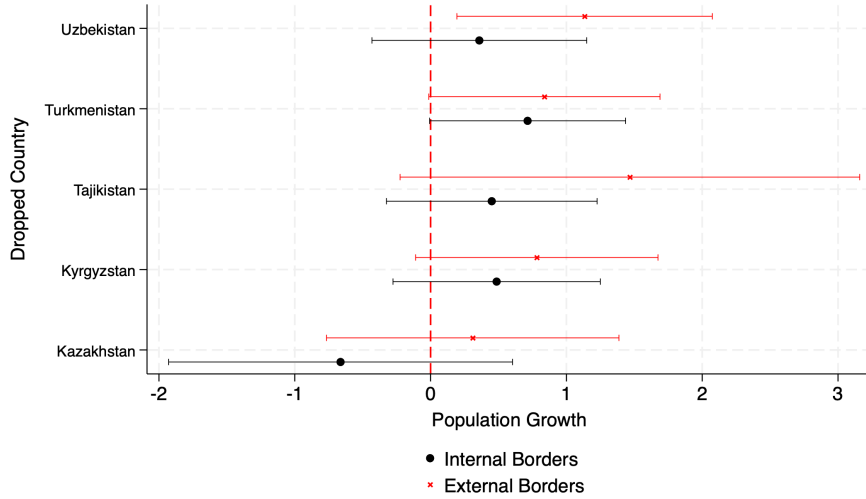
Figure A1: Event-Study Estimates: Sub-Region-by-Period Fixed Effects

Notes: This figure replicates Figure 3 with more demanding fixed effects. Panel A replaces period fixed effects with sub-region-by-period fixed effects, where sub-regions are Baltics, Caucasus, and Slavic Europe plus Moldova. Panel B excludes Turkmenistan, whose census timing (1995, 1999) differs substantially from the other Central Asian countries. All other specifications are identical to Figure 3. Standard errors clustered at the city level; bars show 95% confidence intervals.



(a) Europe

Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. Circles denote IB effects; X markers denote EB effects. Both IB and EB coefficients remain negative across all exclusions. Precision decreases when dropping Russia, which contributes the largest share of cities to the European sample. Bars show 95% confidence intervals.



(b) Central Asia

Notes: Each point represents the estimated coefficient from a specification that drops the labeled country. The EB coefficient remains positive in all five exclusions, though precision varies. The estimate is most attenuated when dropping Kazakhstan, which provides the largest share of interior control cities. The IB coefficient turns negative when Kazakhstan is excluded, again due to the loss in the share of interior control cities. Bars show 95% confidence intervals.

Figure A2: Leave-One-Out Analysis

Table A1: Summary of Census Data

Country	Cities	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Armenia	29	1970	1979	1989	2001	2011	2016
Azerbaijan	43	1970	1979	1989	2002	2011	2015
Belarus	80	1970	1979	1989	1999	2009	2019
Estonia	21	1970	1979	1989	2000	2011	2017
Georgia	41	1970	1979	1989	2002	2014	2020
Kazakhstan	77	1970	1979	1989	1999	2009	2021
Kyrgyzstan	31	1970	1979	1989	1999	2009	2021
Latvia	23	1970	1979	1989	2000	2011	2020
Lithuania	41	1970	1979	1989	2001	2011	2021
Moldova	37	1970	1979	1989	2004	2014	
Russia	702	1970	1979	1989	2002	2010	2021
Tajikistan	29	1970	1979	1989	2000	2010	2021
Turkmenistan	9	1970	1979	1989	1995	1999	
Ukraine	204	1970	1979	1989	2001	2014	2021
Uzbekistan	29	1970	1979	1989	2005	2020	
Total	1396						

Notes: Census years are recorded for each country from 1970 to the most recent available year. Years 1–3 correspond to Soviet-wide censuses (1970, 1979, 1989), which are common across all countries. Years 4–6 correspond to post-independence national censuses, whose timing varies by country.

Table A2: Treatment Variable Statistics by Region

	Mean	[SD]	Min	Max	Treated	Control	Obs.
Panel A: Europe							
<i>Outcome variable:</i>							
Annualized Pop. Growth	0.583	[1.940]	-13.61	27.25	1221 cities		6,004
<i>Main treatment variables:</i>							
Post-1991 (Division)	0.593	[0.491]	0	1	1221	1221	6,004
Internal border (<75 km)	0.281	[0.449]	0	1	353	868	6,004
External border (<75 km)	0.112	[0.316]	0	1	142	1079	6,004
IB $\times$ Division	0.163	[0.369]	0	1	353	1221	6,004
EB $\times$ Division	0.065	[0.246]	0	1	142	1221	6,004
<i>Distance bins (mutually exclusive):</i>							
IB 0–25 km	0.084	[0.277]	0	1	107	775	6,004
IB 25–50 km	0.111	[0.314]	0	1	139	775	6,004
IB 50–75 km	0.086	[0.280]	0	1	107	775	6,004
IB 75–100 km	0.076	[0.264]	0	1	93	775	6,004
EB 0–25 km	0.037	[0.188]	0	1	47	1036	6,004
EB 25–50 km	0.037	[0.190]	0	1	47	1036	6,004
EB 50–75 km	0.038	[0.191]	0	1	48	1036	6,004
EB 75–100 km	0.033	[0.179]	0	1	43	1036	6,004
Panel B: Central Asia							
<i>Outcome variable:</i>							
Annualized Pop. Growth	1.369	[2.301]	-8.71	16.36	175 cities		805
<i>Main treatment variables:</i>							
Post-1991 (Division)	0.565	[0.496]	0	1	175	175	805
Internal border (<75 km)	0.591	[0.492]	0	1	105	70	805
External border (<75 km)	0.076	[0.265]	0	1	15	160	805
IB $\times$ Division	0.330	[0.471]	0	1	105	175	805
EB $\times$ Division	0.039	[0.193]	0	1	15	175	805
<i>Distance bins (mutually exclusive):</i>							
IB 0–25 km	0.323	[0.468]	0	1	58	61	805
IB 25–50 km	0.171	[0.377]	0	1	30	61	805
IB 50–75 km	0.097	[0.296]	0	1	17	61	805
IB 75–100 km	0.050	[0.217]	0	1	9	61	805
EB 0–25 km	0.045	[0.207]	0	1	9	156	805
EB 25–50 km	0.011	[0.105]	0	1	2	156	805
EB 50–75 km	0.020	[0.140]	0	1	4	156	805
EB 75–100 km	0.022	[0.148]	0	1	4	156	805

Notes: All statistics are computed on the estimation panel (city $\times$ census-period observations with non-missing annualized population growth), which matches the regression sample in Tables 1 and A8. All treatment variables are binary indicators. “Treated” and “Control” report the number of unique cities for which the indicator equals one or zero. For time-invariant indicators (e.g., Internal border), the two columns sum to the total number of cities in the region. For interaction terms (e.g., IB $\times$ Division), “Treated” counts border cities, which have the indicator equal to one only in post-1991 periods, while “Control” counts all cities that ever have the indicator equal to zero, including border cities in their pre-1991 periods; the two columns therefore overlap and do not sum to the total. For distance bins, “Control” reports interior cities (>100 km from the relevant border type), the omitted reference category in the gradient regressions (Table A8). Distance bins are mutually exclusive 25 km bands. The outcome variable row reports total unique cities in the estimation sample rather than a treated/control split.

Table A3: Descriptive Statistics by Region

Variable	Group	Mean	SD	Min	Max	Obs.
Europe						
Population	IB=1	76,435	167,631	3000	1,609,959	2,084
	IB=0	116,223	417,391	3059	12,455,682	5,205
1970 Population	IB=1	61,305	136,566	5067	1,222,852	2,118
	IB=0	88,995	299,802	5084	6,941,961	5,208
IB Distance (km)	IB=1	38.48	20.66	0.080	74.52	2,118
	IB=0	460.7	664.9	75.50	5078	5,208
EB Distance (km)	IB=1	311.9	314.0	0.516	1659	2,118
	IB=0	694.5	499.6	0.066	3034	5,208
Large (=1)	IB=1	0.368	0.482	0.000	1.000	2,118
	IB=0	0.566	0.496	0.000	1.000	5,208
Population	EB=1	66,605	140,177	3000	1,201,539	826
	EB=0	109,734	383,409	3059	12,455,682	6,463
1970 Population	EB=1	49,120	103,983	5067	766,705	852
	EB=0	85,184	277,502	5084	6,941,961	6,474
IB Distance (km)	EB=1	341.9	823.6	0.080	3334	852
	EB=0	338.2	555.1	0.259	5078	6,474
EB Distance (km)	EB=1	37.47	22.33	0.066	74.85	852
	EB=0	655.8	471.7	75.36	3034	6,474
Large (=1)	EB=1	0.282	0.450	0.000	1.000	852
	EB=0	0.538	0.499	0.000	1.000	6,474
Central Asia						
Population	IB=1	108,327	252,647	4351	2,571,668	581
	IB=0	81,422	114,737	6010	1,184,469	399
1970 Population	IB=1	70,553	162,565	5053	1,384,509	630
	IB=0	55,322	76,512	6010	523,271	420
IB Distance (km)	IB=1	25.86	19.86	0.655	70.68	630
	IB=0	204.5	108.2	78.52	502	420
EB Distance (km)	IB=1	298.0	280.8	6.290	1540	630
	IB=0	560.8	409.1	2.824	1464	420
Large (=1)	IB=1	0.390	0.488	0.000	1.000	630
	IB=0	0.514	0.500	0.000	1.000	420
Population	EB=1	55,059	107,879	5053	604,700	76
	EB=0	100,930	214,151	4351	2,571,668	904
1970 Population	EB=1	32,674	60,280	5053	253,118	90
	EB=0	67,441	139,683	5239	1,384,509	960
IB Distance (km)	EB=1	134.1	100.0	26.04	382	90
	EB=0	93.88	112.7	0.655	502	960
EB Distance (km)	EB=1	26.35	21.04	2.824	59.52	90
	EB=0	438.5	358.3	80.53	1540	960
Large (=1)	EB=1	0.267	0.445	0.000	1.000	90
	EB=0	0.456	0.498	0.000	1.000	960
Europe						
Log(Lights)	IB=1	7.524	1.735	0.693	11.95	6,072
	IB=0	8.640	1.480	0.693	14.50	13,332
Log(Lights)	EB=1	7.177	1.588	0.693	11.13	2,662
	EB=0	8.468	1.586	0.693	14.50	16,742
Central Asia						
Log(Lights)	IB=1	8.105	1.751	0.693	12.05	1,496
	IB=0	8.134	1.616	3.178	11.54	1,166
Log(Lights)	EB=1	6.945	1.720	3.178	11.50	308
	EB=0	8.271	1.628	0.693	12.05	2,354

Notes: Statistics are reported on the full city $\times$ census-period panel, which includes six periods per city (three pre-1991, three post-1991). Population is the level recorded at each census. 1970 Population is time-invariant but reported for all city-periods. IB and EB Distance measure straight-line kilometers to the nearest internal and external post-Soviet border. Large is a binary indicator for cities above the 1970 sample median population. Log(Lights) is from a separate annual city-year panel (1992–2013) matched to the same city universe; observation counts reflect city $\times$ year. Treatment variable statistics and the outcome variable (annualized population growth) are reported in Table A3, computed on the estimation sample (city-period observations with non-missing growth rates).

Table A4: Border Effects by Type of Exposure

	Dependent Variable: Population Growth			
	(1)	(2)	(3)	(4)
IB only $\times$ Division	−0.358*** (0.113)	−0.172 (0.121)	0.552 (0.395)	0.778** (0.393)
EB only $\times$ Division	−0.441 (0.291)	−0.437 (0.305)	1.356** (0.620)	1.192* (0.629)
Both IB & EB $\times$ Division	−1.099*** (0.266)	−1.047*** (0.244)	0.803 (0.520)	1.078** (0.518)
Both – IB only	−0.740	−0.875	0.251	0.301
p -value	0.007	0.001	0.564	0.492
Both – EB only	−0.657	−0.610	−0.553	−0.113
p -value	0.088	0.105	0.388	0.862
p : effects additive	0.456	0.279	0.143	0.242
Outcome mean (%)	0.583	0.583	1.369	1.309
Sample	Europe	Europe	CA	CA (exc. Turk.)
Cities	1,221	1,221	175	166
Observations	6,004	6,004	805	769
City FE	✓	✓	✓	✓
Period FE	✓		✓	✓
Sub-Region-by-Period FE		✓		

Notes: The dependent variable is annualized city population growth (%). Cities are partitioned into four mutually exclusive categories by exposure: within 75 km of an internal border only (“IB only”), within 75 km of an external border only (“EB only”), within 75 km of both (“Both”), and interior cities beyond 75 km of any border, which form the omitted reference group. “Division” equals one for post-1991 periods. Because the three indicators are mutually exclusive, each coefficient reports the effect of that exposure category relative to interior cities. “Both – IB only” and “Both – EB only” report the incremental effect of adding the second type of border exposure, with p -values from Wald tests of the linear combination. The additivity row reports the p -value from a test that the dual effect equals the sum of the two single-exposure effects; equivalently, it is the p -value on a dual-exposure interaction added to a specification containing both border indicators. Columns (1)–(2) estimate the decomposition for post-Soviet Europe, with Column (2) replacing period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova). Columns (3)–(4) estimate it for Central Asia, with Column (4) excluding Turkmenistan, whose census timing differs from the other Central Asian countries; sub-region-by-period fixed effects are not reported for Central Asia because it constitutes a single sub-region. In post-Soviet Europe, 273 cities are near an internal border only, 62 near an external border only, 80 near both, and 806 are interior. The corresponding counts for Central Asia are 100, 10, 5, and 60. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Border Effects by City Size

	Dependent Variable: Population Growth					
	All		Europe		Central Asia	
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.197 (0.170)	−0.024 (0.105)	−0.447** (0.177)	−0.130 (0.115)	0.923 (0.611)	0.180 (0.405)
EB × Division	−0.567** (0.257)	−0.187 (0.203)	−0.587** (0.273)	−0.200 (0.205)	1.402** (0.629)	0.601 (0.803)
City size	Small	Large	Small	Large	Small	Large
Outcome mean (%)	0.838	0.517	0.752	0.423	1.394	1.337
Treated IB cities	287	171	223	130	64	41
Treated EB cities	113	44	102	40	11	4
Cities	698	698	600	621	98	77
Observations	3,372	3,437	2,919	3,085	453	352
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). “IB” and “EB” denote cities within 75 km of internal and external borders, respectively; “Division” equals one for post-1991 periods. Cities are split at the median 1970 population across the full sample (approximately 28,800 residents), so the cut point is common to both regions rather than computed within region. This absolute threshold is the relevant one for assessing whether the effects are concentrated in the large cities that absorbed the bulk of post-1991 return migration. The pooled cut divides each region close to evenly, placing 49 percent of European and 56 percent of Central Asian cities in the small group.. Because the split partitions treated and control cities together, each column compares border to interior cities within the same size range. Odd-numbered columns report small cities and even-numbered columns large cities, for the pooled sample (Columns 1–2), post-Soviet Europe (Columns 3–4), and Central Asia (Columns 5–6). “Treated IB cities” and “Treated EB cities” report the number of unique cities within 75 km of the respective border type in each estimation sample; these need not sum to the total number of border cities reported elsewhere because a city may lie near both border types. Column (6) contains four treated external border cities and should be read as descriptive. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness Checks

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Europe						
IB × Division	−0.399*** (0.114)	−0.396*** (0.110)	−0.359*** (0.113)	−0.690*** (0.121)	−0.489*** (0.109)	−0.529*** (0.118)
EB × Division	−0.587*** (0.204)	−0.648*** (0.194)	−0.442 (0.291)	−0.412* (0.228)	−0.532*** (0.180)	−0.392** (0.179)
City sample	Pop 1970 <500k	Excl. 1989–99	Excl. both IB&EB	Non-conflict	Excl. Asian RU	Dist <750 km
Outcome mean (%)	0.582	0.786	0.584	0.626	0.597	0.639
Cities	1,191	1,221	1,141	1,049	1,098	771
Observations	5,856	4,783	5,629	5,193	5,389	3,754
Panel B: Central Asia						
IB × Division	0.464 (0.376)	0.466 (0.376)	0.552 (0.395)	0.495 (0.397)	0.569 (0.378)	0.456 (0.450)
EB × Division	0.970** (0.453)	0.755* (0.398)	1.356** (0.620)	1.442* (0.866)	0.938** (0.457)	0.877* (0.485)
City sample	Pop 1970 <500k	Excl. 1989–99	Excl. both IB&EB	Non-conflict	Excl. KZ–RU IB	Dist <750 km
Outcome mean (%)	1.374	1.777	1.349	1.362	1.413	1.412
Cities	172	175	170	155	165	144
Observations	791	630	785	722	755	650
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). Column (1) excludes cities above 500,000 population in 1970, where *propiska* restrictions were most stringent. Column (2) excludes the 1989–1999 intercensal period. Column (3) excludes cities within 75 km of both internal and external borders; because these cities are dropped rather than assigned a separate indicator, its coefficients reproduce the internal-only and external-only estimates in [Table A4](#). Column (4) drops cities near conflict-affected border pairs (Armenia–Azerbaijan, Georgia–Russia, Moldova–Ukraine, Russia–Ukraine, and Kyrgyzstan–Tajikistan for internal; Armenia–Turkey, Afghanistan–Tajikistan for external). Column (5) applies a region-specific restriction. In Panel A it excludes the 123 Russian cities in the Siberian and Far Eastern federal districts, confining the Russian sample to districts west of the Urals and testing whether the classification of Russia with post-Soviet Europe drives the European estimates. In Panel B it excludes the ten Central Asian cities within 75 km of the Kazakhstan–Russia frontier, the one Central Asian internal border that severed a dense industrial corridor. Column (6) restricts the control group to cities within 750 km of any border, to test whether the effects are driven by contrasts with very remote interior cities. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Market Access and Urban Growth: Robustness to Spatial Autocorrelation

	Dependent Variable: Population Growth				
	(1)	(2)	(3)	(4)	(5)
Panel A: Europe					
IB $\times$ Division	−0.402*** (0.106)	−0.402*** (0.122)	−0.402*** (0.136)	−0.402*** (0.146)	−0.402** (0.169)
EB $\times$ Division	−0.595*** (0.184)	−0.595*** (0.203)	−0.595*** (0.209)	−0.595*** (0.203)	−0.595*** (0.198)
Observations	6,004	6,004	6,004	6,004	6,004
Panel B: Central Asia					
IB $\times$ Division	0.471* (0.282)	0.471* (0.279)	0.471* (0.280)	0.471* (0.283)	0.471* (0.275)
EB $\times$ Division	0.969** (0.423)	0.969** (0.424)	0.969** (0.434)	0.969** (0.483)	0.969** (0.470)
Distance cutoff	25 km	50 km	75 km	100 km	200 km
Observations	805	805	805	805	805
City FE	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). Each column reports the baseline specification (city and period fixed effects) with Conley spatial standard errors (Conley, 1999) computed using the `acreg` package (Colella et al., 2019). Columns vary the distance cutoff beyond which spatial correlation is assumed to be zero. Point estimates are identical across columns; only standard errors change. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Distance Gradient: Border Proximity and Urban Growth

	Dependent Variable: Population Growth			
	Europe		Central Asia	
	(1)	(2)	(3)	(4)
IB (0–25 km) × Division	−0.486*** (0.170)	−0.264 (0.171)	0.633 (0.434)	0.886** (0.436)
IB (25–50 km) × Division	−0.363** (0.155)	−0.195 (0.149)	0.488 (0.614)	0.768 (0.631)
IB (50–75 km) × Division	−0.225 (0.196)	−0.118 (0.197)	0.045 (0.574)	0.374 (0.579)
IB (75–100 km) × Division	0.105 (0.168)	0.184 (0.167)	−0.233 (0.475)	0.200 (0.485)
EB (0–25 km) × Division	−1.055*** (0.360)	−1.224*** (0.372)	1.438** (0.626)	1.362** (0.630)
EB (25–50 km) × Division	−0.377 (0.384)	−0.447 (0.363)	0.148 (0.391)	0.440 (0.392)
EB (50–75 km) × Division	−0.485* (0.258)	−0.561** (0.254)	0.755 (0.636)	0.084 (0.337)
EB (75–100 km) × Division	−0.835*** (0.206)	−0.925*** (0.224)	0.953 (1.126)	0.257 (0.982)
Outcome mean (%)	0.583	0.583	1.369	1.309
Cities	1,221	1,221	175	166
Observations	6,004	6,004	805	769
City FE	✓	✓	✓	✓
Period FE	✓		✓	✓
Sub-Region-by-Period FE		✓		

Notes: The dependent variable is annualized city population growth (%). Each row reports the coefficient on the interaction between a distance-bin indicator and a post-1991 indicator. The omitted reference category is cities beyond 100 km from any border. Column (1) is the baseline for Europe with city and period fixed effects. Column (2) replaces period fixed effects with sub-region-by-period fixed effects (Baltics, Caucasus, Slavic Europe plus Moldova). Column (3) is the baseline for Central Asia. Column (4) excludes Turkmenistan. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Summary Statistics: Sectoral Composition (1989)

	IB Regression Sample					EB Regression Sample				
	Mean	Med.	SD	Min	Max	Mean	Med.	SD	Min	Max
Panel A: Europe										
Manufacturing share	0.834	0.925	0.234	0	1	0.841	0.924	0.224	0	1
Agriculture share	0.026	0	0.088	0	1	0.022	0	0.068	0	0.813
Mining share	0.074	0	0.204	0	1	0.069	0	0.196	0	1
Total empl. (000s)	17.3	5.381	40.6	0.022	785	17.2	5.653	41.5	0.022	785
(total)	1021					848				
of which: IB cities	290 (0.284)					62 (0.073)				
of which: EB cities	62 (0.061)					117 (0.138)				
of which: Interior	731 (0.716)					731 (0.862)				
Heterogeneity diagnostics (treated cities):										
Manuf. quartile (Q1/Q2/Q3/Q4)	249(77)/258(59)/255(64)/259(90)					203(31)/221(22)/220(29)/204(35)				
Manuf. median split (High/Low)	514(154) / 507(136)					424(64) / 424(53)				
Agric. median split (High/Low)	494(127) / 527(163)					414(47) / 434(70)				
Mining median split (High/Low)	294(90) / 727(200)					240(36) / 608(81)				
Panel B: Central Asia										
Manufacturing share	0.753	0.849	0.281	0	1	0.686	0.755	0.286	0	1
Agriculture share	0.036	0	0.111	0	1	0.074	0.001	0.17	0	1
Mining share	0.111	0	0.253	0	1	0.102	0	0.225	0	0.93
Total empl. (000s)	12.1	4.728	20.5	0.026	143	10.1	3.03	18.3	0.118	118
(total)	141					61				
of which: IB cities	91 (0.645)					3 (0.049)				
of which: EB cities	3 (0.021)					11 (0.18)				
of which: Interior	50 (0.355)					50 (0.82)				
Heterogeneity diagnostics (treated cities):										
Manuf. median split (High/Low)	72(55) / 69(36)					19(2) / 42(9)				
Agric. median split (High/Low)	69(45) / 72(46)					32(8) / 29(3)				
Mining median split (High/Low)	45(27) / 96(64)					19(1) / 42(10)				

Notes: Sectoral shares are calculated as sector employment divided by total city employment in 1989. The IB regression sample includes all cities except those near only external borders, matching internal border specifications (Tables 1–2). The EB regression sample includes all cities except those near only internal borders. “Of which” rows report the count and share of each city type; IB and EB counts need not sum to the total because some cities fall within 75 km of both border types. In the heterogeneity diagnostics, numbers in parentheses indicate treated (border) cities; the remainder are interior controls. “Manuf. quartile” reports the Q1/Q2/Q3/Q4 distribution across manufacturing quartiles used in Table 2, Columns (1)–(2). “Median split (High/Low)” reports cell sizes for specifications in Tables 2 and A10. In both regions, the median agriculture and mining shares are zero, so “High” captures cities with any positive employment in that sector.

Table A10: Urban Growth and Sectoral Composition: Additional Median Splits

	Dependent Variable: Population Growth			
	Internal Border		External Border	
	(1) Agric.	(2) Mining	(3) Agric.	(4) Mining
Panel A: Europe				
Border × Div. × High Agric.	−0.307* (0.165)		−0.584 (0.387)	
Border × Div. × Low Agric.	−0.446** (0.181)		−0.758** (0.295)	
Border × Div. × High Mining		0.059 (0.196)		−0.170 (0.314)
Border × Div. × Low Mining		−0.578*** (0.155)		−0.915*** (0.305)
High Agric. × Div.	0.005 (0.126)		0.005 (0.126)	
High Mining × Div.		−0.215 (0.146)		−0.215 (0.146)
<i>p</i> -value: High = Low	0.570	0.011	0.720	0.090
Outcome mean (%)	0.527	0.527	0.578	0.578
Cities	1,021	1,021	848	848
Observations	5,033	5,033	4,206	4,206
Panel B: Central Asia				
	(1) Manuf.	(2) Mining	(3) Manuf.	(4) Mining
Border × Div. × High Manuf.	0.197 (0.529)		1.483 (1.291)	
Border × Div. × Low Manuf.	0.499 (0.604)		1.479* (0.778)	
Border × Div. × High Mining		1.470* (0.859)		1.702** (0.840)
Border × Div. × Low Mining		−0.133 (0.445)		0.915 (0.652)
High Manuf. × Div.	0.476 (0.711)		0.475 (0.717)	
High Mining × Div.		−1.577* (0.894)		−1.577* (0.902)
<i>p</i> -value: High = Low	0.705	0.099	0.997	0.453
Outcome mean (%)	1.362	1.362	1.328	1.328
Cities	141	141	61	61
Observations	652	652	284	284
City FE	✓	✓	✓	✓
Period FE	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). Cities are split at the within-region median of the relevant 1989 employment share; “High” denotes cities above the median and “Low” denotes cities at or below the median. For agriculture and mining in Europe, where the median share is zero, the split compares cities with any positive employment in that sector to those without. Panel A reports Europe, excluding manufacturing (shown in the main table as quartiles). Panel B reports Central Asia, excluding agriculture (shown in the main table as a median split). “Border” refers to cities within 75 km of an internal border in Columns (1)–(2) and an external border in Columns (3)–(4). “Div.” equals one for post-1991 periods. The Central Asian external border sample contains 61 clusters, of which 11 are near external borders; sectoral splits yield small cells, so Columns (3)–(4) of Panel B should be interpreted with caution. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Covariate Balance in Matched Samples

Matching variable	Treated (Border)					Control (Interior)				
	Mean	SD	Min	Max	<i>N</i>	Mean	SD	Min	Max	<i>N</i>
Panel A: Europe										
<i>Internal border matching:</i>										
1970 Population	61,465	136,889	5,235	1,222,852	352	70,598	140,246	5,226	1,170,133	415
1989 Ind. Employment	15,564	31,924	54.0	250,404	289	15,634	30,281	24.0	274,489	399
1989 Ind. Turnover	340,085	644,501	70.0	4,758,280	290	370,136	696,305	64.0	4,796,282	379
1989 Manufacturing	216,068	463,404	0.000	3,781,983	290	259,122	514,307	0.000	3,728,328	383
1989 SIC	13,427	6,547	2,730	32,720	279	12,564	6,432	2,730	32,720	719
Military count	8.084	17.8	1.000	160.0	179	18.6	29.5	1.000	208.0	94
Military size	17.6	38.0	2.000	339.0	179	40.6	63.0	2.000	443.0	94
<i>External border matching:</i>										
1970 Population	49,433	104,595	5,235	766,705	141	53,785	108,825	5,226	770,905	238
1989 Ind. Employment	12,137	24,526	54.0	161,117	115	13,034	25,561	59.0	158,143	199
1989 Ind. Turnover	256,859	504,018	70.0	3,176,533	117	254,788	505,249	64.0	3,107,729	207
1989 Manufacturing	159,088	326,866	0.000	2,117,026	117	168,095	339,349	0.000	2,161,119	198
1989 SIC	13,673	6,316	2,730	27,520	115	12,196	6,151	2,730	27,520	826
Military count	7.758	15.4	1.000	105.0	66	14.7	28.1	1.000	166.0	49
Military size	16.5	32.5	2.000	218.0	66	31.7	60.1	2.000	360.0	49
Panel B: Central Asia										
<i>Internal border matching:</i>										
1970 Population	54,241	76,263	6,197	430,618	97	57,006	81,603	6,010	523,271	62
1989 Ind. Employment	11,542	16,064	129.0	87,890	86	11,075	19,623	118.0	117,556	52
1989 Ind. Turnover	312,309	448,680	1,330	2,403,345	88	456,404	672,555	954.0	2,731,450	48
1989 Manufacturing	191,509	340,011	0.000	1,825,853	91	187,719	319,809	0.000	1,414,383	49
1989 SIC	10,820	6,123	2,730	23,000	69	11,694	6,408	2,730	23,000	42
Military count	3.209	4.877	1.000	28.0	43	3.200	2.484	1.000	10.0	15
Military size	7.047	10.2	2.000	58.0	43	7.267	5.788	2.000	23.0	15
<i>External border matching:</i>										
1970 Population	34,647	63,901	6,010	253,118	14	41,022	72,711	5,879	266,815	21
1989 Ind. Employment	3,771	6,317	163.0	21,598	11	4,166	6,473	147.0	22,022	20
1989 Ind. Turnover	98,293	130,617	2,523	382,986	11	104,737	131,277	2,080	386,449	20
1989 Manufacturing	51,211	86,545	0.000	281,351	11	57,188	88,302	0.000	302,282	20
1989 SIC	8,169	4,720	2,730	20,110	9	9,170	5,908	2,730	20,110	48
Military count	1.333	0.577	1.000	2.000	3	1.500	0.577	1.000	2.000	4
Military size	2.667	1.155	2.000	4.000	3	3.000	1.155	2.000	4.000	4

Notes: Each row reports summary statistics of the matching variable for treated (border) and control (interior) cities in the corresponding matched sample. Each matching exercise is independent: the row for “1970 Population” under internal border matching reports balance in the sample matched on 1970 population (Table 3, Column 1 for Europe; Table A11, Column 1 for Central Asia). Columns (1)–(4) and (6) of the matching tables use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. The 1:2 ratio balances two considerations: using more than one neighbor improves efficiency in the small Central Asian samples, while restricting to two avoids drawing poor matches from distant regions of the covariate distribution. With replacement and a fixed neighbor count, exact balance on the matching variable is not guaranteed, particularly when the treated and control distributions have different shapes or when a few control cities serve as neighbors for many treated cities. Column (5) uses exact matching on each city’s primary SIC code, retaining all border and interior cities that shared the same dominant industrial classification in 1989. Military matching uses two covariates (count and workforce size of defense enterprises closed 1989–1991), so both are reported. Statistics are computed on the cross-section (one observation per city).

Table A12: Robustness to Matching on Pre-1991 Characteristics (Central Asia)

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Internal Borders						
IB × Division	0.597 (0.424)	0.511 (0.481)	0.607 (0.507)	0.733 (0.468)	0.026 (0.408)	1.424* (0.711)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	1.332	1.344	1.382	1.389	1.548	1.593
Cities	151	131	129	133	106	58
Observations	697	603	595	614	483	266
Adj. R ²	0.227	0.289	0.291	0.328	0.292	0.455
Panel B: External Borders						
EB × Division	3.007* (1.696)	1.086 (1.040)	0.736 (1.113)	2.699 (2.139)	−0.533 (0.651)	3.819* (1.450)
Matching on	1970 Pop.	1989 Ind. Empl.	1989 Ind. Turn.	1989 Manuf.	1989 SIC	Military
Outcome mean (%)	1.525	1.553	1.515	1.486	1.746	2.294
Cities	21	16	23	19	25	5
Observations	91	69	102	83	110	22
Adj. R ²	0.229	0.192	0.083	0.012	0.306	0.451
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: Each column re-estimates the baseline specification on a matched subsample of treated border cities and their nearest neighbors. Columns (1)–(4) and (6) use 1:2 nearest-neighbor propensity score matching with replacement on the city cross-section. Column (5) uses exact matching on each city’s primary SIC code, restricting the sample to border and interior cities that shared the same dominant industrial classification in 1989. In Central Asia, matched samples are small, particularly for external borders (Panel B), where the treated sample contains only 11–15 cities depending on covariate availability. The negative EB coefficient in Column (5) of Panel B (−0.533) rests on just 25 cities and likely reflects the composition of the few SIC strata that contain both treated and control cities rather than a substantive finding. Column (6) of Panel B retains just 5 cities, so this estimate should not be over-interpreted. “Military” matches jointly on the count and total workforce of defense enterprises that closed between 1989 and 1991. The dependent variable is annualized city population growth (%). “Division” equals one for post-1991 periods. All specifications include city and period fixed effects. Standard errors clustered at the city level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Excluding Science Cities and Closed Cities

	Dependent Variable: Population Growth					
	(1)	(2)	(3)	(4)	(5)	(6)
IB × Division	−0.395*** (0.110)	−0.392*** (0.110)	−0.379*** (0.111)	−0.365*** (0.112)	−0.361*** (0.113)	0.479 (0.384)
EB × Division	−0.609*** (0.200)	−0.607*** (0.200)	−0.600*** (0.200)	−0.592*** (0.201)	−0.593*** (0.202)	0.833* (0.459)
Outcome mean (%)	0.553	0.554	0.547	0.540	0.543	1.355
Sample	Europe	Europe	Europe	Europe	Europe	CA
Science cities excl.	3 km	5 km	10 km	20 km	20 km	
Closed cities excl.					✓	✓
Cities	1,170	1,166	1,150	1,119	1,112	162
Observations	5,749	5,729	5,649	5,494	5,460	745
City FE	✓	✓	✓	✓	✓	✓
Period FE	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is annualized city population growth (%). “IB” and “EB” denote cities within 75 km of internal and external borders, respectively; “Division” equals one for post-1991 periods. Each column re-estimates the baseline specification after excluding cities that hosted Soviet place-based strategic investment, to test whether the post-1991 divergence reflects the collapse of that investment rather than border hardening. Columns (1)–(4) exclude cities matched to the 95 Science Cities identified by [Schweiger et al. \(2022\)](#) at successively wider radii; the list covers the territory of modern Russia only, so these exclusions bear on the European sample. Matching is by coordinates, with the reported radius applied to the distance between each sample city and the nearest Science City. Only one lies within 75 km of an internal border. Widening the radius beyond 10 km removes adjacent towns rather than additional Science Cities, so Column (4) should be read as a neighbourhood exclusion. Columns (5)–(6) additionally exclude 20 closed or restricted-access cities outside Russia, hand-coded from secondary sources: 7 in post-Soviet Europe (Paldiski, Sillamäe, Liepāja, Sevastopol, Zhovti Vody, Kamianske, and Dnipro) and 13 in Central Asia (Baikonur, Stepnogorsk, Aktau, Ust-Kamenogorsk, Chkalovsk, Taboshar, Khaidarkan, Kadamjay, Kara-Balta, Mailuu-Suu, Zarafshon, Chirchik, and Navoiy). Because closed cities were largely omitted from the published Soviet statistical sources on which the population panel is built, several of the most restricted sites (Sarov, Ozyorsk, Seversk, Novouralsk, Lesnoy, Kurchatov, Sniečkus, and Slavutych) do not appear in the sample at all. Standard errors clustered at the city level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A1 - Data Cleaning

Night Lights Data

Several steps were undertaken to transform pixel-level lights data into city-level data following Storeygard (2016). First, 30 satellite-years of night lights data were merged into a single binary grid, indicating whether a pixel was lit above a digital number threshold of 20 in at least one year.²⁰ These ever-lit areas were then converted into polygons by aggregating contiguous lit pixels. For each polygon and year, I summed the raw digital number values (ranging from 0 to 63) of all constituent pixels to obtain a continuous measure of total light intensity. The regression outcome is the log of this summed intensity, which captures variation in economic activity across cities and over time.

I matched polygons to known cities using census population data with latitude and longitude coordinates from Brinkhoff (2021). Polygons that did not correspond to a known city were excluded; these likely represent forest fires, sensor noise not flagged by the satellite algorithm, or smaller towns and large villages, and account for 10 to 20 percent of total digital numbers in the 15-country sample.²¹

Additionally, I removed lights from gas flares, as identified by Elvidge et al. (2009). To further minimize errors, I clipped the nighttime lights raster to the national boundaries of FSU countries. Without this step, light emitted by cities in neighboring non-FSU countries (e.g., Poland, Finland, China) could bleed into FSU border areas through the well-documented blooming effect in DMSP data, artificially inflating light intensity for FSU cities near external borders. The night lights dataset is annual from 1992 to 2013, enabling analysis only for the period following the dissolution.²²

²⁰The threshold of 20 is applied only at the polygon construction stage to define city boundaries by identifying reliably lit urban areas; it does not affect the outcome variable used in regressions. Li et al. (2020) show that thresholds of 20 to 30 effectively eliminate blooming artifacts and sensor noise around urban peripheries while retaining reliably lit areas.

²¹An alternative approach would use all lit pixels within a fixed buffer around each city centroid, without a brightness threshold. This would avoid potential censoring of newly developed areas at urban peripheries but would also introduce noise from non-urban light sources. The polygon-based approach used here ties the nightlights outcome directly to known urban footprints identified in the census data, ensuring that the measured light corresponds to the same cities whose population growth is analyzed in the main results.

²²Gibson et al. (2021) highlight key issues with DMSP night lights data, such as blurring and top-coding, which can distort economic measurements. They particularly note that these errors lead to poor predictions of economic activity in low-density rural areas. However, the study focuses on whole cities with populations of at least five thousand people in 1970, where these data provide more reliable insights. Additionally, the results are robust to dropping large cities such as Moscow, where top-coding may be an issue.

Appendix B - Effects by Individual Border Pair

To further probe how the effect differs for each border pair, I estimate the following three-way interaction model separately for internal and external borders:

$$y_{ct} = \sum_{j=1}^J \beta_j (\text{border}_c \times \text{division}_t \times \mathbf{1}\{\text{borderpair}_c = j\}) + \alpha_c + d_t + \epsilon_{ct}, \quad (6)$$

where border_c indicates proximity to internal or external borders, $\mathbf{1}\{\text{borderpair}_c = j\}$ is an indicator equal to one if city c 's nearest border belongs to border pair j and zero otherwise, and J is the total number of border pairs. City fixed effects α_c absorb all time-invariant city characteristics, including border-pair membership, while year fixed effects d_t control for common census year shocks. This model allows isolation of how the Soviet Union's dissolution uniquely influenced population growth around distinct borders.

Figure B1 presents results for internal border pairs. In Europe, the majority of border pairs exhibit negative coefficients, though magnitudes and precision vary considerably with sample size. The largest and most precisely estimated declines are at Latvia–Lithuania (-1.70 , significant at the 1% level, 31 cities), Moldova–Ukraine (-1.61 , significant at the 1% level, 46 cities), Belarus–Lithuania (-1.43 , significant at the 1% level, 22 cities), and Belarus–Ukraine (-1.07 , significant at the 1% level, 26 cities). Armenia–Azerbaijan shows a large negative effect (-1.76 , significant at the 1% level), though the Nagorno-Karabakh conflict complicates interpretation. Belarus–Russia (-0.35 , significant at the 5% level) and Lithuania–Russia (-1.09 , significant at the 1% level) are also negative, while Russia–Ukraine is modestly positive (0.30 , significant at the 5% level), potentially reflecting the continued dominance of bilateral Russia–Ukraine trade despite border frictions. This uplift requires specific context: rather than cleanly severing ties, the economies of eastern Ukraine and western Russia remained deeply intertwined throughout the 1990s, particularly in heavy industry, metallurgy, and energy transit. The imposition of a national boundary transformed these previously frictionless neighboring cities into essential customs, logistics, and transit hubs for the largest bilateral trade corridor in the former Soviet Union. Several pairs with very few cities (Belarus–Latvia with 1 city, Azerbaijan–Russia with 7 cities, Latvia–Russia with 4 cities) produce extreme or imprecise estimates that should not be over-interpreted.

In Central Asia, six of the seven internal border pairs show positive coefficients, with Kazakhstan–Russia the exception. Tajikistan–Uzbekistan (1.05 , $p = 0.09$) and Turkmenistan–Uzbekistan (1.08 , $p = 0.10$) are the most precisely estimated, though the intervals remain wide throughout. The point estimates are positive for six of the seven pairs, but the confidence intervals are wide and none excludes zero at conventional levels, so the pair-level evidence does not overturn the aggregate Central Asian internal border null in **Table 1**. Both are consistent with the limited disruption expected from enforcing borders across weakly integrated economies.

Figure B2 presents results for external border pairs. In Europe, the largest and most precisely estimated negative effects are at Moldova–Romania (-1.78 , significant at the 1% level, 26 cities) and Lithuania–Poland (-1.97 , significant at the 1% level, 9 cities), both former COMECON frontiers where trade reoriented toward the EU. Belarus–Poland is also negative (-0.58 , significant at the 5% level). Several Caucasian border pairs show mixed results: Armenia–Turkey (-0.58) and Georgia–Turkey (-0.35) are negative but imprecise, consistent with the limited commercial potential of these frontiers noted in Section 2. Poland–Russia shows a notable positive effect (1.01 , significant at the 1% level), likely reflecting Kaliningrad's unique economic arrangements as a Russian exclave with special trade provisions. Border pairs with very few cities (Hungary–Ukraine with 2 cities, Finland–Russia with 4 cities) are imprecise.

In Central Asia, all external border pairs show positive coefficients. Iran–Turkmenistan has the largest effect (3.68 , significant at the 1% level), followed by Afghanistan–Tajikistan

(1.55, significant at the 5% level). China–Kazakhstan (0.68, $p = 0.13$) is positive but imprecise, reflecting the single city (Zharkent) near this border. The uniformly positive direction across geographically distinct frontiers reinforces the interpretation that border opening generated market access gains in Central Asia. To reiterate, individual estimates are imprecise for pairs with few cities, so the value of this exercise lies in the broad pattern rather than any single coefficient.

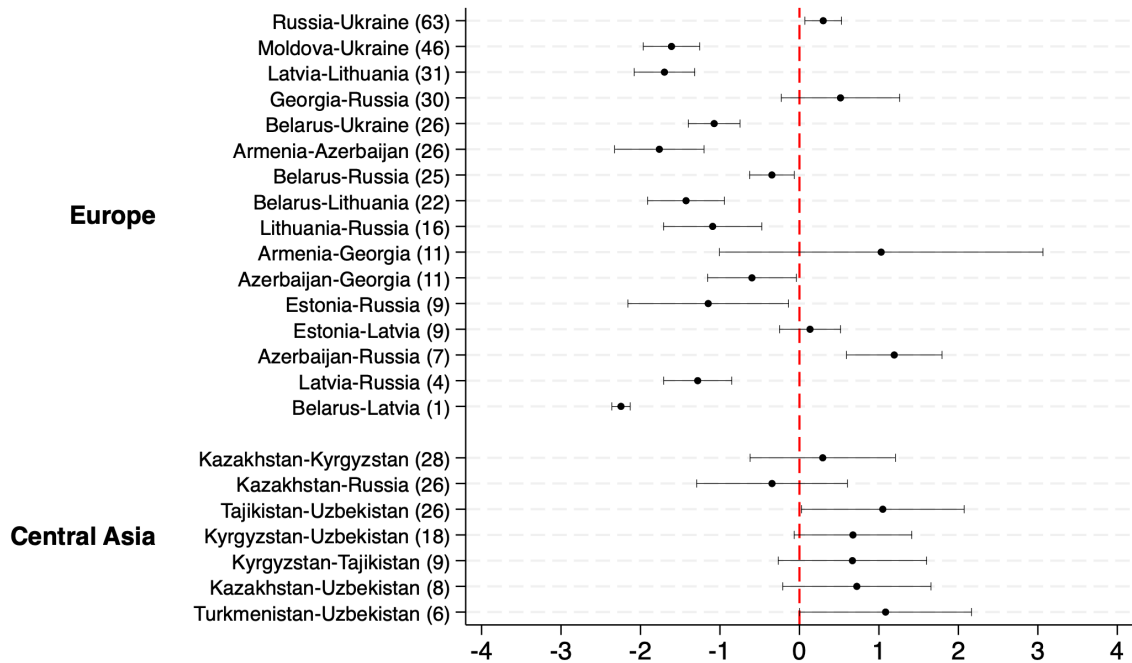


Figure B1: Treatment Effects by Internal Border Pair

Notes: Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated internal border pair, estimated from equation 6. Separate regressions are run for Europe and Central Asia, with interior cities in each region as controls. Numbers in parentheses indicate the number of border cities per pair. Border pairs sorted by city count within each region. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.

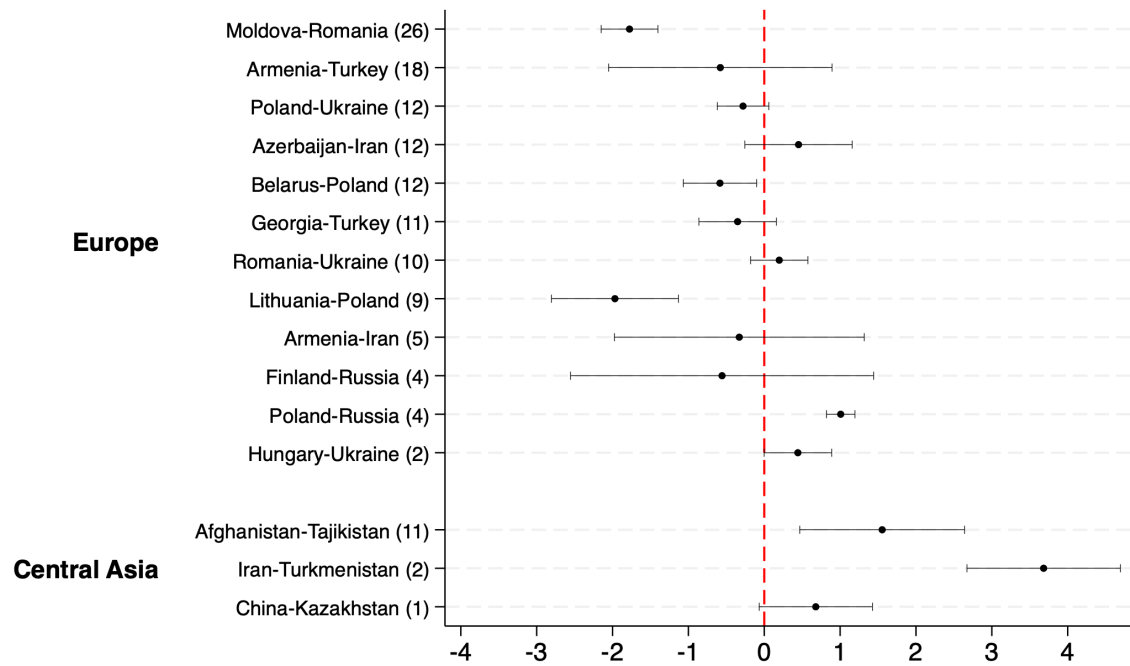


Figure B2: Treatment Effects by External Border Pair

Notes: Each coefficient represents the post-1991 effect on annualized population growth for cities within 75 km of the indicated external border pair, estimated from equation 6. Separate regressions are run for Europe and Central Asia, with interior cities in each region as controls. Numbers in parentheses indicate the number of border cities per pair. All specifications include city, period, and border-pair fixed effects. Standard errors clustered at the city level. Bars show 90% confidence intervals.

Appendix C - Effects on Economic Activity Proxied by Night Lights

In this section, I examine whether the city-level population results for Europe and Central Asia also manifest in nighttime lights data, a widely used proxy for local economic activity. Although nighttime light data is only available from 1992 onward, I exploit this limitation by treating 1992, the first full year after the Soviet dissolution, as the omitted baseline. I estimate separate event-study specifications for internal and external border cities, comparing changes in log night lights relative to interior cities over the period 1993–2013.

$$\log(\text{lights})_{ct} = \sum_{\tau=1993}^{2013} \beta_{\tau} (\text{border}_c \times \mathbf{1}\{t = \tau\}) + \alpha_c + d_t + \epsilon_{ct}, \quad (7)$$

where $\log(\text{lights})_{ct}$ is the log of summed lights in city c in year t , border_c is a dummy equal to one if a city is within 75 km of a border and zero otherwise, and $\mathbf{1}\{t = \tau\}$ is an indicator for year τ . The year 1992 serves as the omitted baseline, so each coefficient β_{τ} captures the difference in log night lights between border and interior cities in year τ relative to 1992. City fixed effects α_c absorb time-invariant city characteristics, while year fixed effects d_t control for common annual shocks. The specification is estimated separately for internal border cities (IB) and external border cities (EB), with the two sets of coefficients plotted together in Figure C1. While this approach does not allow for direct observation of pre-1992 trends, it reveals important information about the relative trajectories of economic activity following the dissolution.

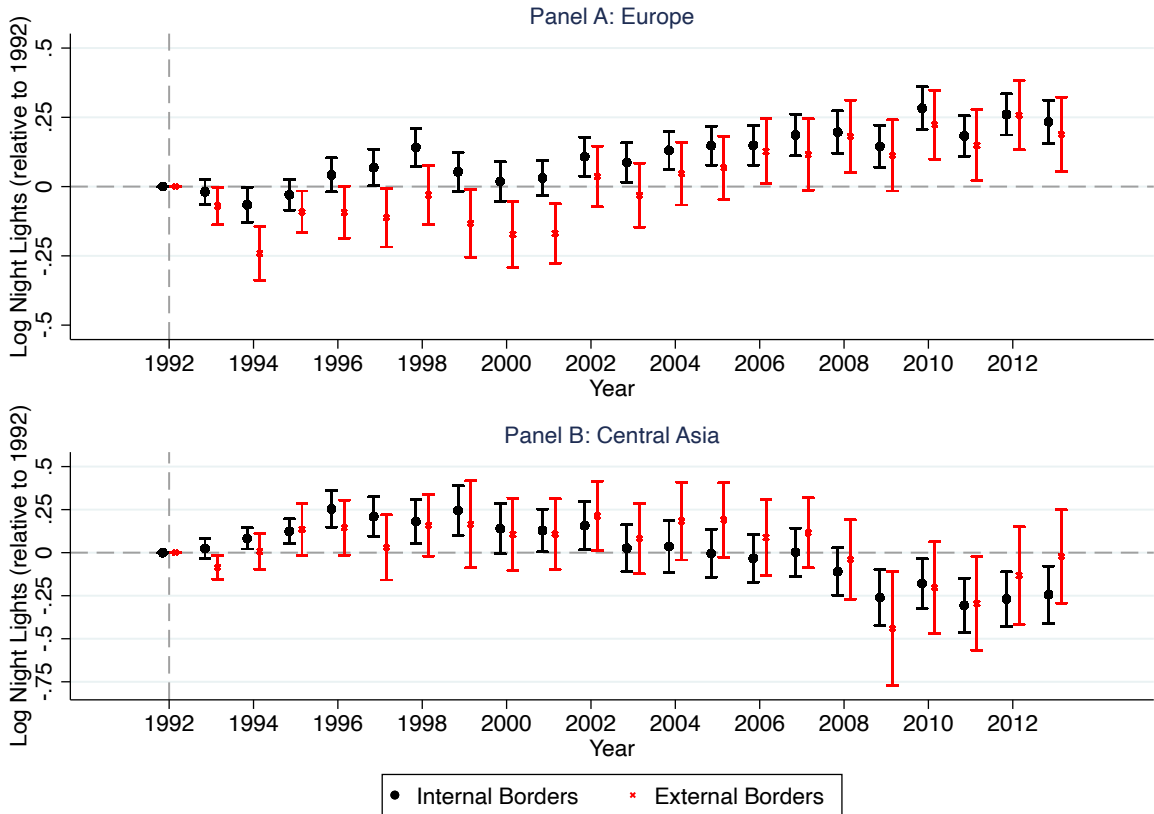


Figure C1: Event-study Estimates of Nighttime Light Intensity relative to 1992

Notes: This figure plots coefficients from event-study regressions of log nighttime light intensity on border proximity interacted with year indicators, relative to the 1992 baseline. Black circles denote internal borders; red X's denote external borders. Panel A shows Europe; Panel B shows Central Asia. Because nightlights data begin in 1992, no pre-treatment trends can be assessed. All regressions include city, year, and border-pair fixed effects. Standard errors clustered at the city level; bars show 90% confidence intervals.

Figure C1 presents results for Europe (Panel A) and Central Asia (Panel B). Two important caveats apply when interpreting these results alongside the main population findings. First, nighttime lights data begin in 1992, so there is no pre-treatment period to validate parallel trends. Second, the population regressions estimate annualized growth rates (a flow measure), while the nightlights specification estimates log levels relative to a 1992 baseline. A negative coefficient in the population regression implies continuously diverging population levels, whereas the nightlights coefficients capture the cumulative deviation in economic activity relative to 1992. These differences in outcome measurement mean the two sets of results need not produce identical time-series patterns.

In Europe (Panel A), internal border cities show an initial relative decline in 1994, followed by a rebound that produces positive deviations from the mid-1990s onward. External border cities follow a similar but delayed trajectory, with coefficients near zero through the early 2000s before turning consistently positive after 2005. At first glance, these positive deviations appear to contradict the population results, which show sustained negative effects. However, the two findings are not necessarily inconsistent. The population results capture relative *growth rates*: border cities grew more slowly than interior cities in every post-1991 period. The nightlights results capture relative *levels* of economic activity measured from a 1992 baseline that is itself already depressed by the initial collapse. The positive nightlights deviations after 2002 may reflect catch-up in economic activity per capita as border city populations shrank while economic output partially stabilized, or they may reflect the normalization of border institutions, EU accession of neighboring countries, and increased cross-border cooperation. The nightlights evidence is therefore best interpreted as showing that the acute economic disruption at borders was concentrated in the 1990s, with some recovery in relative economic activity thereafter, even as population continued to decline in relative terms.

In Central Asia (Panel B), internal border cities show positive and increasing deviations through the late 1990s, consistent with initially limited disruption from the enforcement of inter-republican borders. However, these positive trajectories flatten and turn negative from the mid-2000s onward, potentially reflecting increasing state control, higher tariffs, and tighter border enforcement in countries such as Uzbekistan and Turkmenistan. External border cities present a pattern that contrasts sharply with their European counterparts. Whereas European external border cities showed near-zero or negative deviations through the early 2000s, Central Asian external border cities display positive coefficients from the mid-1990s through the early 2000s, with the 2002 coefficient marginally significant. This divergence is consistent with the core finding that the same geopolitical rupture generated opposite market access shocks across regions: European external border cities lost access to reorienting COMECON partners while Central Asian external border cities gained access to previously sealed frontiers. The Central Asian external border gains attenuate and turn negative from 2009 onward, coinciding with the formalization of border regimes and reduced scope for informal shuttle trade. The imprecision of many individual-year coefficients reflects the small sample of external border cities (57 clusters) and the inherent noisiness of DMSP data in sparsely populated regions.

Overall, the nightlights evidence provides partial corroboration of the population findings. The cross-regional contrast in external border trajectories, negative in Europe and positive in Central Asia during the first post-collapse decade, mirrors the opposing signs documented in the population regressions and reinforces the market access interpretation. The two outcome measures capture different dimensions of the border shock, however, and should not be expected to produce identical time-series patterns.